CHAPTER – 01
LIVING WORD

1. Non-living objects
   (1) do not grow.
   (2) do not reproduce.
   (3) undergo metabolism.
   (4) can respond to stimuli.
2. Which of the following match is incorrect?
   (1) Fungi Spore formation
   (2) Hydra - Budding
   (3) Planaria – Regeneration
   (4) Yeast - Conidia
3. Which of the following possess the ability to reproduce?
   (1) Mules and hinny
   (2) Worker honey bees
   (3) Drone honey bees
   (4) Infertile human couples
4. An isolated metabolic reaction(s) outside the body of an organism, performed in a test tube is
   (1) either living or non-living.
   (2) neither living nor non-living.
   (3) always living.
   (4) always non-living.
5. Cell division occurs in plants and in animals.
   (1) continuously; only up to a certain age
   (2) only up to a certain age; continuously
   (3) continuously; never
   (4) once; twice
6. Organisms that can respond to stimuli are
   (1) only eukaryotes.
   (2) only prokaryotes.
   (3) Both (1) and (2).
   (4) those with a well-developed nervous system.
7. Reproduction is synonymous with growth/cell division in
   (a) Bacteria. (b) Hydra. (c) Planaria. (d) Unicellular algae..
   Choose the correct option.
   (1) (a), (c) and (e) (2) (a), (b) and (d) (3) (a), (d) and (e) (4) All of these
8. Which of the following does not easily multiply by fragmentation?
   (1) Fungi (2) Filamentous algae (3) Amoeba (4) Protonema of mosses
9. can indicate categories at very different levels.
   (1) Taxa (2) Rank (3) Key (4) Catalogue
10. Consider the following characteristics:
    (a) Increase in mass
    (b) Differentiation
    (c) Increase in number of individuals
    (d) Response to stimuli
    Which two points are known as the twin characteristics of growth?
    (1) (a) and (b) (2) (a) and (d) (3) (b) and (c) (4) (a) and (c)
11. Which set of organisms multiply by fragmentation?
    (1) Earthworm, Amoeba, fungi
    (2) Earthworm, fungi, bacteria
    (3) Fungi, filamentous algae, protonema of mosses
    (4) Amoeba, Hydra, bacteria
12. Metabolic reactions take place in
    (1) isolated cell-free systems.
    (2) living systems.
    (3) Both (1) and (2).
    (4) Either (1) or (2).
13. Which of the features given below are general characteristics of life?
    (a) Growth
    (b) Reproduction
    (c) Response to stimuli
    (d) Metabolism
    (e) Cellular organization
    (1) All except (c) (2) All of these (3) (b), (c) and (d) (4) All except (e)
14. Properties of tissues are
    (1) present in the constituent cells.
    (2) due to similar cells in them.

(3) due to their similar origin.
(4) a result of interactions among the constituent cells.

15. Select the correct statement.
(1) Only living organisms grow.
(2) Plants grow only up to a certain age.
(3) The growth in living organisms is from inside.
(4) All of these.

16. Select the incorrect pair.
(1) Hydra – Budding
(2) Flatworm - Regeneration
(3) Amoeba – Fragmentation
(4) Yeast – Budding

17. Reproduction is synonymous with growth in
(1) most of the fungi and Planaria.
(2) desmids, diatoms and protozoans.
(3) cyanobacteria, fungi and mosses.
(4) mosses, algae and Hydra.

18. All living organisms have following defining features:
(1) growth and reproduction
(2) metabolism and reproduction
(3) consciousness and metabolism
(4) cellular organisation and reproduction

19. Which of the following sets of organisms can reproduce to produce fertile offspring?
(1) Worker bees, blue whales and elephants
(2) Queen bee, Drosophila and yeast
(3) Mules, sharks and kangaroos
(4) Sterile human couples, hinny and donkeys.

20. Fill in the blanks and select the option with all correct answers.
(a) Increase in body is considered as growth.
(b) In living organisms, growth is from .
(c) A organism does not grow.
(d) Unicellular organisms grow by .
(1) (a) - mass; (c) – unicellular
(2) (b) - outside; (c) - dead
(3) (a) - mass; (b) – inside
(4) (c) - multicellular; (d) - asexual reproduction

21. Select the incorrect option.
(1) All living phenomena are due to underlying interactions.
(2) Properties of tissues arise as a result of interactions among the constituent cells.
(3) Living organisms are self-replicating, evolving and self-regulating interactive systems capable of responding to external stimuli.
(4) All members of kingdom Animalia possess a specific characteristic called self-consciousness.

22. The patients lying in coma in hospitals virtually supported by machines
(1) has no self-consciousness.
(2) undergoes metabolism.
(3) can reproduce at a slow pace.
(4) Both (1) and (2).

23. Match the entities in Column-I with their mode of reproduction in Column-II.

Column-I	**Column-II**
(a) Planaria	(i) Binary fission
(b) Fungi	(ii) Asexual spores
(c) Yeast	(iii) Budding
(d) Hydra	(iv) True regeneration
(e) Amoeba	(v) Fragmentation
(f) Species Plantarum and Systema Naturae	(vi) Linnaeus

(1) (a) - (i); b-(ii); (c) - (iii); (d) - (iv); (e) - (vi); (f) - (v)
(2) (a) - (iv); b-(ii); (c) - (v); (d) - (iii); (e) - (i); (f) - (vi)

(3) (a) - (v); b - (ii); (c) - (iv); (d) - (iii); (e) - (i); (f) - (vi)
(4) (a) - (ii); b - (iii); (c) - (i); (d) - (iv); (e) - (ii); (f) - (vi)

24. Which one of the following aspects is an exclusive characteristic of living things?
(1) Increase in mass from inside only.
(2) Isolated metabolic reactions occur in vitro.
(3) Perception of events happening in the environment and their memory.
(4) Increase in mass by accumulation of material both on surface as well as internally.

25. Which of the following is incorrect for reproduction?
(1) Unicellular organisms reproduce by cell-division.
(2) Reproduction is a characteristic of all living organisms.
(3) In unicellular organisms, reproduction and growth are linked together.
(4) Non-living objects are incapable of reproducing.

26. Mark the incorrect statement with respect to metabolism.
(1) Microbes exhibit the metabolism.
(2) It is the property of all living forms.
(3) The metabolic reactions can be demonstrated in vitro.
(4) It is not a defining feature of life forms.

27. Find out the incorrect statement with respect to reproduction.
(1) In unicellular organisms, growth and reproduction are two different processes.
(2) Reproduction is of two types in animals: asexual and sexual.
(3) Asexual reproduction does not produce variations.
(4) Many living organisms which do not reproduce are mules, sterile worker bees and infertile human couples.

28. Select the incorrect statement for consciousness as one of the characteristic of living organisms.
(1) It is the most technically complicated process.
(2) It is present in all organisms whether unicellular or multicellular.
(3) It is an obvious feature not found in prokaryotes.
(4) It is regarded as a defining feature of living organisms.

29. Consider the following statements:
(a) Increase in mass and increase in number of individuals are twin characteristics of growth.
(b) A multicellular organism grows by cell division.
(c) In plants, the growth is seen up to a certain age.
(d) In majority of higher animals and plants growth and reproduction are mutually inclusive events.
(e) The growth exhibited by non-living objects is by accumulation of materials on the surface.
Choose the correct option.
(1) (a) and (c) (2) (c) and (d) (3) (a), (b) and (c) (4) (a) and (b)

30. From the following statements, select those that are true for reproduction.
(a) It is not an all-exclusive defining characteristic of living organisms.
(b) It is not an all-inclusive defining characteristic of living organisms.
(c) It is not an all-exclusive defining characteristic of plants and fungi only.
(d) Photoperiod affects reproduction in seasonal plant breeders only.
(e) Photoperiod affects reproduction in seasonal plants and animals breeders.
(f) Photoperiod has no role in reproduction.
(1) (b) and (e) (2) (c) and (d) (3) (d) and (f) (4) All of these

31. Choose the incorrect statements from the following:
(a) Growth cannot be taken as a defining property of living organism.
(b) Dead organism does not grow.
(c) Reproduction cannot be an all-inclusive defining characteristic of living organisms.

(d) No non-living object is capable of replicating itself.
(e) Metabolism in a test tube is non-living.
(f) Metabolism is a defining feature of all living organisms.
(1) (a) and (c) (2) All except (e) (3) All except (c) (4) All of these

32. Select the correct statement for growth as one of the characteristic of living organisms.
(1) Growth by increase in mass is a defining property of prokaryotic organisms only.
(2) Protoplasmic growth is feature of higher organisms only.
(3) Intrinsic growth is a characteristic of all living organisms.
(4) Growth can be extrinsic or intrinsic for multicellular organisms.

33. Read the following statements and select correct option for all living organisms.
(a) Reversible increase in body mass by accumulation of symplasmic materials from outside.
(b) Intrinsic growth.
(c) Growth and reproduction are mutually inclusive events in multicellular organisms.
(d) All living objects are capable of replicating by themselves.
(e) Cellular organisation of the body is defining property.
(1) (a), (c) and (e) (2) (b), (c) and (d) (3) (b) and (e) (4) (a), (b), (d) and (e)

34. Fill in the blank. All living organisms - present, past and future, are linked to one another by the sharing of common but to varying degrees.
(1) genetic material (2) chemical composition
(3) cell structure (4) metabolism

35. Mayer's biological concept of species is mainly based on
(1) morphological traits. (2) reproductive isolation.
(3) modes of reproduction. (4) morphology and reproduction.

36. Which of the following is incorrect with respect to species?
(1) A group of individual organisms with fundamental similarities.
(2) Two different species breed together to produce fertile offspring.
(3) Human beings belong to the species sapiens.
(4) Panthera has many specific epithet as tigris, leo and pardus.

37. Select the correct statement with respect to biodiversity.
(1) There are around $1.7-1.8$ million species on the earth.
(2) The number of mammals on the Earth is maximum as compared to all animal species.
(3) Out of the known species the plant alone are 1.2 million species.
(4) Every year just 15,000 species are taxonomically discovered.

38. Which of the following statements is correct?
(1) Species diversity, in general, increases from poles to the equator.
(2) Conventional taxonomic methods are equally suitable for higher plants and microorganisms.
(3) India's share of global species diversity is about 18%.
(4) There are about 25,000 species of plants known in India.

39. Select the wrong statement.
(1) The number of species that are known and described range between 1.7-1.8 million.
(2) The scientific names ensure that each organism has only one name.
(3) The first word in a biological name represents the specific epithet while the second component denotes the genus.
(4) Biological names are generally in Latin and written in italics.

40. In nomenclature,
(1) both genus and species are printed in italics.
(2) genus and species are always of same name.

(3) both in genus and species, the first letter is capital.
(4) genus is written after the species.

41. The disadvantage of using common names for species is that
(1) the names may change.
(2) one name does not apply universally.
(3) one species may have several common names and one common name may be applied to two species.
(4) All of the above.

42. In referring to an organism in writing, such as in a newspaper, textbook or lab report, which of these rules should be followed?
(a) Underline or italicize genus
(b) Underline or italicize species
(c) First letter of species should be uppercase
(d) First letter of genus should be uppercase
(1) All except (c) (2) All except (d) (3) All except (a) (4) All except (b)

43. In trinomial nomenclature,
(1) the third term in animals represents the strain.
(2) the third term in plants denotes the variety.
(3) when species and sub-species are same, those names are called tautonyms.
(4) the third term in micro-organisms denotes the sub-species.

44. The third name in trinomial nomenclature in animals represents
(1) genus. (2) species. (3) sub-genus. (4) sub-species.

45. When the specific epithet exactly repeats generic name, it is called
(1) basinym. (2) synonym. (3) homonym. (4) tautonym.

46. Read the statements given below and identify the incorrect statement.
(1) Scientific names are used all over the world.
(2) Scientific names indicate relationship between species.
(3) Scientific names favour multiple naming for the same kind of an organism.
(4) Scientific names are often descriptive and tell us some important character of an organism.

47. The biological concept of species was given by
(1) John Ray. (2) Linnaeus. (3) Aristole. (4) Ernst Mayr.

48. ICBN is
(1) International Code of Biological Naming.
(2) International Code of Botanical Nomenclature.
(3) International Class of Biological Nomenclature.
(4) International Classification of Biological Nomenclature.

49. Scientific names of plants are based on principles and criteria agreed by and are given in
(1) IUCN. (2) ICZN. (3) ICBN. (4) ICPN.

50. Identify the generic name from the given scientific names.
Panthera leo, Canis familiaris, Rosa indica
(1) Leo, Canis, indica (2) Panthera, Canis, Rosa
(3) Leo, familiaris, indica (4) Panthera, Rosa, familiaris

51. According to binomial nomenclature, every living organism has
(1) two scientific name with single component.
(2) one scientific name with two components.
(3) two names, one Latin and other common.
(4) one common name with three components.

52. Which of the following is incorrect with respect to binomial nomenclature?
(1) Biological names are generally in Latin.
(2) The first word in a biological name represents the genus.
(3) Biological names are printed in italics.
(4) The first word of the genus starts with a small letter.

53. Which one of the following statements given below is not included in universal rules of nomenclature?
(1) Generic names and specific epithet should be in Latin words.
(2) Generic name is immediately followed by name of taxonomists who described it first.
(3) Generic name must begin with capital letter.
(4) All letters of the specific name must be small.

54. Select the mismatched pair.
(1) Order Primata - Gibbon, monkey, gorilla
(2) Order Carnivora - Tiger, cat, dog
(3) Order Polymoniales - Solanaceae and Liliaceae families
(4) Order Carnivora - Felidae and Canidae families

55. A taxon is a
(1) group of related families. (2) group of related species.
(3) type of living organisms. (4) taxonomic group of any ranking.

56. Identify the correct sequence of taxonomic categories.
(1) Species - Order - Kingdom – Phylum (2) Species - Family - Genus - Class
(3) Genus - Species - Order – Phylum (4) Species - Genus - Order – Phylum

57. Two species which are morphologically almost identical but they do not interbreed are called
(1) Monotypic species. (2) Polytypic species.
(3) Evolutionary species. (4) Sibling species.

58. The practical purpose of classification of living organisms is to
(1) name the living organisms.
(2) explain the origin of living organisms.
(3) trace the evolution of living organisms.
(4) facilitate identification of unknown organisms.

59. When organism is in same division but is not in same order, the taxonomic term is called
(1) class. (2) genus.1 (3) species. (4) family.

60. Which of the following taxa covers a least number of organisms?
(1) Class (2) Order (3) Genus (4) Phylum

61. Linnaean system of classification was based on
(1) cytology. (2) morphology. (3) ecology. (4) embryology.

62. Scientific study of diversity of organisms and their evolutionary relationships is called
(1) anatomy. (2) taxonomy. (3) systematics. (4) morphology.

63. Which of the following are not the processes that are basic to taxonomy?
(1) Characterisation (2) Systematics (3) Identification (4) Nomenclature

64. Which one of the following has least similar characteristics?
(1) Family (2) Class (3) Genus (4) Species

65. Which one of the following is an odd category?
(1) Species (2) Class (3) Phylum (4) Sub family

66. Who wrote Species Plantarium and provided a basis for classification of plants?
(1) Carolus Linnaeus (2) Charles Darwin
(3) Robert Hooke (4) Leeuwenhoek

67. Arrangement of taxa is termed as

(1) analogy. (2) taxonomy. (3) hierarchy. (4) homology.

68. Each unit of a category of classification is termed as

(1) taxon. (2) order. (3) series. (4) trophic level.

69. The correct sequence of taxa is:

(1) Class - Order - Family - Tribe - Genus - Species.

(2) Class - Order - Tribe - Family - Genus - Species.

(3) Phylum - Order-Class - Tribe - Genus - Species.

(4) Phylum - Tribe - Class - Order - Genus - Species.

70. The basic unit of taxonomy/biological classification is

(1) species. (2) genus. (3) phylum. (4) order.

71. When organism is in the same class but is not in the same family, the taxonomic term is called

(1) order. (2) genus. (3) species. (4) family.

72. Which of the following taxa covers a greater number of organisms?

(1) Class (2) Order (3) Genus (4) Phylum

73. Animals are classified into hierarchical groups. In which one of the following, the largest number of species is found?

(1) Genus (2) Order (3) Family (4) Phylum

74. Class is the category of taxonomy which includes related

(1) families. (2) orders. (3) genera. (4) species.

75. In taxonomic hierarchy, which of the following group of taxa will have less number of similarities as compared to other?

(1) Solanaceae, Convolvulaceae and Poaceae (2) Polymoniales, Poales and Sapindales

(3) Solanum, Petunia and Atropa (4) Leopard, tiger and lion

76. Match Column-I with Column-II and select the correct option.

Column-I	**Column-II**
(a) Homo sapiens	(i) Primata
(b) Mangifera indica	(ii) Poales
(c) Musca domestica	(iii) Sapindales
(d) Triticum aestivum	(iv) Diptera

(1) (a) - (i); (b) - (iii); (c) - (ii); (d) - (iv) (2) (a) - (iii); (b) - (ii); (c) - (iv); (d) - (i)

(3) (a) - (i); (b) - (iii); (c) - (iv); (d) - (ii) (4) (a) - (i); (b) - (iv); (c) - (iii); (d) - (ii)

77. Fill in the blanks (a) and (b) using the correct option. Kingdom → Phylum → (a) → Order → (b)

(1) (a) - Genus; (b) – Species (2) (a) - Family; (b) - Class

(3) (a) - Class; (b) – Family (4) (a) - Species; (b) – Division

78. Taxonomy deals with

(1) the development of zoological parks.

(2) the study of kinds and diversity of microorganisms only.

(3) the evolutionary relationships between organisms.

(4) the classification of diverse organisms in different taxa.

79. Which of the following suffix is used for category order?

(1) -phyta (2) -phyceae (3) -ales (4) -aceae

80. The given table gives the classification of a wheat plant.

(a)Kingdom Plantae (b)Division Angiospermae

(c)Monocotyledonae (d)Poales

(e) Family

Fill in the blanks using the correct option.

(1) (a) - Genus; (b) - Class; (c) – Poaceae
(2) (a) - Class; (b) - Order; (c) - Poaceae
(3) (a) - Genus; (b) - Class; (c) – Solanaceae
(4) (a) - Class; (b) - Order; (c) – Solanaceae

81. The correct sequence of taxonomic study of a newly discovered organism is
(1) firstclassification thenidentification, nomenclature and characterization.
(2) first identification then classification and then characterization and nomenclature.
(3) first nomenclature then characterization, identification and classification.
(4) first characterization then identification and nomenclature and then classification.

82. The equivalent rank of Carnivora in taxonomic categories of man and housefly is respectively
(1) Homo and Musca.
(2) Hominidae and Muscidae.
(3) Mammalia and Insecta.
(4) Primata and Diptera.

83. A group of related genera in taxonomic hierarchy has/is
(1) still more number of common characters than lower categories.
(2) less number of general characters w.r.t. lowest category.
(3) characterised on the basis of vegetative and reproductive features.
(4) less number of similarities in comparison to upper categories.

84. Taxa like sweet potato and mango are placed in taxonomic hierarchy in the
(1) different orders but same class and phylum.
(2) different orders but same class and division.
(3) same class and order but different family.
(4) different class but same division.

85. Select correct statement for genus.
(1) Genus is a group of individual organisms with fundamental similarities.
(2) Genus has one or more than one species representing monotypic or polytypic species condition.
(3) Genus has more characters in common in comparison to species of other genera.
(4) Genus is an aggregate of related and unrelated species.

86. Fill in the blanks by choosing the correct option
(a) Each may have one or more than one specific epithets representing different organisms.
(b) Taxonomic studies consider a group of individual organisms with fundamental similarities as a .
(c) In case of plants, classes with a few similar characters are assigned to a higher category called .
(d) being a higher category, is the assemblage of families which exhibit a few similar characters.

	(a)	(b)	(c)	(d)
(1)	species	order	phylum	Division
(2)	genus	species	division	Order
(3)	class	genus	species	Phylum
(4)	species	genus	division	Tribe

87. Which of the following statements about classification is not true?
(a) Members of a family are less similar than members of an included genus.
(b) An order has more members than the number of members in an included genus.
(c) Families have more members than phyla.
(d) Members of a family share a common ancestor in the more distant past than members of an

included genus.

(e) The number of species in a taxon depends on their relative degree of similarity.

(1) Only (c) (2) Only (d) (3) Only (e) (4) None of these

88. Biosystematics aims at

(1) the classification of organisms based on their evolutionary history and establishing their phylogeny studies.

(2) identification and arrangement of organisms on the basis of their cytological characteristics.

(3) the classification of organisms based on broad morphological characters.

(4) delimiting various taxa of organisms and establishing their relationships.

89. Modern taxonomic studies do not deal with

(a) external structure. (b) structure of cell.

(c) internal structure. (d) phylogeny.

(e) ontogeny. (f) nutrition.

(1) (d) and (f) (2) (b) and (c) (3) (a) and (c) (4) (c) and (e)

90. 90. Consider the following taxa and select how many of them do not belong to the lowest category of taxonomic hierarchy?

91. Chilli, Lion, Dicots, Animals, Mango, Sweet potato, Polymoniales, Cat

(1) Seven (2) Six (3) Five (4) Three

91. Place of keeping and studying dry plants is

(1) arboretum. (2) museum.q (3) botanical garden. (4) herbarium.

92. An important function of botanical garden is

(1) providing beautiful area for recreation.

(2) one can observe tropical plants over there.

(3) they allow ex situ conservation of germplasm.

(4) to allow reproduction freely with each other and form seeds.

93. National Botanical Institute is located in

(1) Kolkata. (2) Mumbai. (3) Chennai. (4) Lucknow.

94. The Indian Botanical garden is located at

(1) Chennai. (2) Kolkata. (3) Lucknow. (4) Dehradun.

95. Which taxonomic aid gives comprehensive account of complete compiled information of a genus or family at a particular time?

(1) Flora (2) Monograph (3) Dictionary (4) Herbarium

96. Place of keeping and studying live plant species is

(1) aquarium. (2) museum. (3) botanical garden. (4) herbarium.

97. Lloyd Botanical Institute is located in

(1) Kolkata. (2) Mumbai. (3) Darjeeling. (4) Lucknow.

98. The Royal Botanical garden is located at

(1) Chile. (2) Kew. (3) UAE. (4) St. Louis.

99. Which one of the following is a taxonomical aid for identification of plants and animals based on similarities and dissimilarities?

(1) Flora (2) Keys (3) Manuals (4) Catalogues

100. The largest collection of herbarium in India is

(1) Blatter Herbarium, Mumbai. (2) Central Circle Herbarium, Allahabad.

(3) Central National Herbarium, Kolkata. (4) Southern Circle Herbarium, Coimbatore.

101. National Botanical Research Institute consists of

(1) dried and preserved plant specimens only.

(2) collection of preserved plant and animal specimens.

(3) flora, manuals and monographs only.
(4) collection of living plants for reference.

102. In museums, animals are
(1) kept and preserved in containers or jars.
(2) preserved in boxes after killing.
(3) stuffed and then preserved.
(4) All of the above.

103. Key is
(1) a form of herbaria.
(2) a type of educational institute.
(3) a taxonomical aid used for identifying various organisms.
(4) a taxonomic category.

104. In zoological parks, animals are
(1) kept and preserved in containers or jars.
(2) preserved in boxes after killing.
(3) kept in protected environments under human care.
(4) stuffed and then preserved.

105. For identifying organisms through key usually
(1) two contrasting characters are used.
(2) one similar character is studied.
(3) two or more similar characters are used.
(4) only one statement called lead is used.

106. The largest Zoological Park in the world is located at
(1) S. America (Rio Jani).
(2) England (Cruger).
(3) Australia (Queen's Land).
(4) S. Africa (Victoria).

107. Dead specimens of plants and animals are found in
(1) herbarium and museums
(2) botanical garden and zoological parks
(3) museums and aquariums
(4) zoological parks and gene banks

108. Which one of the following is not a correct statement?
(1) Herbarium houses are used to dry, press and preserve plant specimens.
(2) Botanical gardens have collection of living plants for reference.
(3) A museum has collection of photographs of plants and animals only.
(4) Key is a taxonomic aid for identification of specimens.

109. Which one of the following is not a correct statement?
(1) Keys are analytical in nature.
(2) Zoological parks have collection of living animals for reference.
(3) A monograph gives a detailed account of all the plant species found in a particular geographical area.
(4) Carolous Linnaeus is regarded as father of nomenclature.

110. Herbarium sheets are arranged according to the system of classification and should have information about
(1) time and place of collection, English, local and botanical names, phylum, collector's name.
(2) date and time of collection, English, local and botanical names, class, collector's name.
(3) date and place of collection, English, local and botanical names, order, collector's name.
(4) date and place of collection, English, local and botanical names, family, collector's name.

111. Match Column-I with Column-II and choose the correct option.
\begin{tabular}{|l|c|}
\hline \multicolumn{1}{|c|}{ Column-I } & \multicolumn{1}{c|}{ Column-II } \\
\hline (a) Ex-situ conservation & \begin{tabular}{l}
(i) Central National \\
Herbarium
\end{tabular} \\

\hline (b) Quick referral system & (ii) Museum \\
\hline \begin{tabular}{l}
(c) Preserved plants and \\
animals
\end{tabular} & (iii) Flora \\
\hline \begin{tabular}{l}
(d) Actual account of hab- \\
itat and distribution of \\
plants in a given area
\end{tabular} & \begin{tabular}{l}
(iv) Royal Botanical \\
Garden, Kew
\end{tabular} \\
\hline
\end{tabular}

(1) (a) - (ii); (b) - (iii); (c) - (iv); (d) - (i) (2) (a) - (i); (b) - (iv); (c) - (ii); (d) - (iii)
(3) (a) - (iv); (b) - (i); (c) - (iii); (d) - (ii) (4) (a) - (iv); (b) - (i); (c) - (ii); (d) - (iii)

112. Among the given taxonomic aids how many are associated with preservation of specimens?
Key, Flora, Museum, Botanical Garden, Catalogue, Herbarium, Zoological Park, Monograph
(1) One (2) Three (3) Two (4) Four

113. Live specimens of plants and animals are found respectively in
(1) herbarium and museums. (2) botanical garden and zoological parks.
(3) museums and herbarium. (4) zoological parks and botanical gardens.

114. Ex-situ conservation strategies of live specimens where each specimen is labelled indicating its scientific name and its family, are
(1) museum. (2) botanical gardens.
(3) monograph. (4) more than one option is correct.

115. Which of the following statement(s) about taxonomical aids is/are true?
(a) Keys are used to identify plants and animals based on similarities and dissimilarities.
(b) Flora contains the account of habitat and distribution of plants in a given area.
(c) Flora provide an index to the plant species found in a particular area.
(d) Monographs provide information for identifying the species found in an area.
(1) (a) and (b) (2) All except (d) (3) (a) and (d) (4) Only (a)

116. The taxonomical aid used for identification of plants and animals based on the similarities and dissimilarities
(1) also show actual account of habitat.
(2) maintains collections of preserved plant and animal specimens.
(3) is generally analytical in nature.
(4) is associated with in-situ conservation strategies.

117. Select correct statement w.r.t. taxonomical aid which is generally analytical in nature.
(1) It serves as quick referral system.
(2) Specimens are preserved in solutions.
(3) It represents the choice made between two opposite options.
(4) It is exclusively used for identification of animals.

118. The taxonomical aids which serves as quick referral systems in taxonomical studies is linked with
(1) preservation of flora and fauna.
(2) labelling that provide information about scientific name and family name of plants and animals.

(3) ex-situ conservation of organisms.
(4) drying, pressing and preservation methods.
Miscellaneous

119. Which of the following is not included under in situ conservation?
(1) National park (2) Botanical garden (3) Sanctuary (4) Biosphere reserve

120. Match Column-I with Column-II and choose the correct option.

Column-I	Column-II
(a) Binomial nomenclature	(i) Carolus Linnaeus
(b) Generic name	(ii) Muscidae
(c) Family	(iii) Panthera
(d) Systema naturae	

(1) (a) - (i); (b) - (iii); (c) - (iii); (d) - (ii) (2) (a) - (i); (b) - (iii); (c) - (ii); (d) - (i)
(3) (a) - (ii); (b) - (i); (c) - (i); (d) - (iii) (4) (a) - (iii); (b) - (i); (c) - (ii); (d) - (i)

121. Consider the following statements:
(a) Increase in mass and increase in number of individuals are twin characteristics of growth.
(b) A multicellular organism grows by cell division.
(c) In plants, the growth is seen up to a certain age.
(d) In majority of higher animals and plants growth and reproduction are mutually inclusive events.
(e) The growth exhibited by non-living objects is by accumulation of materials on the surface.
Choose the correct option.
(1) (a) and (c) (2) (c) and (d) (3) (a), (b) and (c) (4) (a) and (b)

122. Non-living objects
(1) do not grow. (2) do not reproduce.
(3) undergo metabolism. (4) can respond to stimuli.

123. Which of the following match is incorrect?
(1) Fungi - Spore formation (2) Hydra - Budding
(3) Planaria – Regeneration (4) Yeast - Conidia

124. Fill in the blanks and select the option with all correct answers.
(a) Increase in body   is considered as growth.
(b) In living organisms, growth is from   .
(c) A   organism does not grow.   .
(d) Unicellular organisms grow by   .
(1) (a) - mass; (c) – unicellular (2) (b) - outside; (c) - dead
(3) (a) - mass; (b) – inside (4) (c) - multicellular; (d) - asexual reproduction

125. Which of the following possess the ability to reproduce?
(1) Mules and hinny (2) Worker honey bees
(3) Drone honey bees (4) Infertile human couples

126. An isolated metabolic reactions) outside the body of an organism, performed in a test tube is
(1) either living or non-living.q (2) neither living nor non-living.
(3) always living. ` (4) always non-living.

127. Select the incorrect option.
(1) All living phenomena are due to underlying interactions.
(2) Properties of tissues arise as a result of interactions among the constituent cells.
(3) Living organisms are self-replicating, evolving and self-regulating interactive systems capable of

responding to external stimuli.

(4) All members of kingdom Animalia possess a specific characteristic called self-consciousness.

128. The patients lying in coma in hospitals virtually supported by machines

(1) has no self-consciousness. (2) undergoes metabolism.

(3) can reproduce at a slow pace. (4) Both (1) and (2)..

129. Cell division occurs in plants and in animals.

(1) continuously; only up to a certain age (2) only up to a certain age; continuously

(3) continuously; never (4) once; twice

130. Match the entities in Column-I with their mode of reproduction in Column-II.

Column-I	Column-II
(a) Planaria	(i) Binary fission
(b) Fungi	(ii) Asexual spores
(c) Yeast	(iii) Budding
(d) Hydra	(iv) True regeneration
(e) Amoeba	(v) Fragmentation
(f) Species Plantarum and Systema Naturae	(vi) Linnaeus

(l) (a) - (i); b-(ii); (c) - (iii); (d) - (iv); (e) - (vi); (f) - (v)

(2) (a) - (iv); b-(ii); (c) - (v); (d) - (iii); (e) - (i); (f) - (vi)

(3) (a) - (v); b - (ii); (c) - (iv); (d) - (iii); (e) - (i); (f) - (vi)

(4) (a) - (ii); b-(iii); (c) - (i); (d) - (iv); (e) - (ii); (f) - (vi)

131. Organisms that can respond to stimuli are

(1) only eukaryotes.

(2) only prokaryotes.

(3) Both (1) and (2).

(4) those with a well-developed nervous system.

132. From the following statements, select those that are true for reproduction.

(a) It is not an all-exclusive defining characteristic of living organisms.

(b) It is not an all-inclusive defining characteristic of living organisms.

(c) It is not an all-exclusive defining characteristic of plants and fungi only.

(d) Photoperiod affects reproduction in seasonal plant breeders only.

(e) Photoperiod affects reproduction in seasonal plants and animals breeders.

(f) Photoperiod has no role in reproduction.

(l) (b) and (e) (2) (c) and (d) (3) (d) and (f) (4) All of these.

133. 13. Reproduction is synonymous with growth/cell division in

(a) Bacteria. (b) Hydra.

(c) Planaria. (d) Unicellular algae.

(e) Amoeba.

Choose the correct option.

(1) (a), (c) and (e) (2) (a), (b) and (d) (3) (a), (d) and (e) (4) All of these

131. Which one of the following aspects is an exclusive characteristic of living things?
(1) Increase in mass from inside only.
(2) Isolated metabolic reactions occur in vitro.
(3) Perception of events happening in the environment and their memory.
(4) Increase in mass by accumulation of material both on surface as well as internally.

132. Which of the following does not easily multiply by fragmentation?
(1) Fungi (2) Filamentous algae
(3) Amoeba (4) Protonema of mosses

133. can indicate categories at very different levels.
(1) Taxa (2) Rank (3) Key (4) Catalogue

134. Consider the following characteristics:
(a) Increase in mass (b) Differentiation
(c) Increase in number of individuals (d) Response to stimuli
Which two points are known as the twin characteristics of growth?
(1) (a) and (b) (2) (a) and (d) (3) (b) and (c) (4) (a) and (c)

131. 18. Which set of organisms multiply by fragmentation?
(1) Earthworm, Amoeba, fungi
(2) Earthworm, fungi, bacteria
(3) Fungi, filamentous algae, protonema of mosses
(4) Amoeba, Hydra, bacteria

132. Metabolic reactions take place in
(1) isolated cell-free systems. (2) living systems.
(3) Both (1) and (2). (4) Either (1) or (2).

133. Which of the features given below are general characteristics of life?
(a) Growth (b) Reproduction
(c) Response to stimuli (d) Metabolism
(e) Cellular organization
(1) All except (c) (2) All of these (3) (b), (c) and (d) (4) All except (e)

134. Choose the incorrect statements from the following:
(a) Growth cannot be taken as a defining property of living organism.
(b) Dead organism does not grow.
(c) Reproduction cannot be an all-inclusive defining characteristic of living organisms.
(d) No non-living object is capable of replicating itself.
(e) Metabolism in a test tube is non-living.
(f) Metabolism is a defining feature of all living organisms.
(1) (a) and (c) (2) All except (e) (3) All except (c) (4) All of these

135. Properties of tissues are
(1) present in the constituent cells.
(2) due to similar cells in them.
(3) due to their similar origin.
(4) a result of interactions among the constituent cells.

136. Select the correct statement.
(1) Only living organisms grow. (2) Plants grow only up to a certain age.
(3) The growth in living organisms is from inside. (4) All of these.

137. Select the incorrect pair.
(1) Hydra – Budding (2) Flatworm - Regeneration
(3) Amoeba – Fragmentation (4) Yeast – Budding

138. Which of the following is incorrect for reproduction?
(1) Unicellular organisms reproduce by cell-division.
(2) Reproduction is a characteristic of all living organisms.
(3) In unicellular organisms, reproduction and growth are linked together.
(4) Non-living objects are incapable of reproducing.

139. Mark the incorrect statement with respect to metabolism.
(1) Microbes exhibit the metabolism.
(2) It is the property of all living forms.
(3) The metabolic reactions can be demonstrated in vitro.
(4) It is not a defining feature of life forms.

140. Select the correct statement for growth as one of the characteristic of living organisms.
(1) Growth by increase in mass is a defining property of prokaryotic organisms only.
(2) Protoplasmic growth is feature of higher organisms only.
(3) Intrinsic growth is a characteristic of all living organisms.
(4) Growth can be extrinsic or intrinsic for multicellular organisms.

141. Reproduction is synonymous with growth In
(1) most of the fungi and Planarla.
(2) desmids, diatoms and protozoans.
(3) cyanobacteria, fungl and mosses.
(4) mosses, algae and Hydra.

142. Find out the Incorrect statement with respect to repro. duction.
(1) In unicellular organisms, growth and reproduction are two different processes.
(2) Reproduction is of two types in animals: asexual and sexual.
(3) Asexual reproduction does not produce variations.
(4) Many living organisms which do not reproduce are mules, sterile worker bees and Infertile human couples.

143. Select the incorrect statement for consciousness as one of the characteristic of living organisms.
(1) It is the most technically complicated process.
(2) It is present in all organisms whether unicellular or multicellular.
(3) It is an obvious feature not found in prokaryotes.
(4) It is regarded as a defining feature of living organisms.

144. All living organisms have following defining features:
(1) growth and reproduction
(2) metabolism and reproduction
(3) consciousness and metabolism
(4) cellular organisation and reproduction

145. Which of the following sets of organisms can reproduce to produce fertile offspring?
(1) Worker bees, blue whales and elephants
(2) Queen bee, Drosophila and yeast
(3) Mules, sharks and kangaroos
(4) Sterile human couples, hinny and donkeys.

Diversity in the Living World and Species Concept

146. All living organisms - present, past and future, are linked to one another by the sharing of common but to varying degrees.
(1) genetic material
(2) chemical composition
(3) cell structure
(4) metabolism

147. Mayer's biological concept of species is mainly based on
(1) morphological traits.
(2) reproductive isolation.
(3) modes of reproduction.
(4) morphology and reproduction.

148. Which of the following is incorrect with respect to species?
(1) A group of individual organisms with fundamental similarities.
(2) Two different species breed together to produce fertile offspring.

(3) Human beings belong to the species sapiens.
(4) Panthera has many specific epithet as tigris, leo and pardus.

149. Select the correct statement with respect to biodiversity.
(1) There are around 1.7-1.8 million species on the earth.
(2) The number of mammals on the Earth is maximum as compared to all animal species.
(3) Out of the known species the plant alone are 1.2 million species.
(4) Every year just 15,000 species are taxonomically discovered.

150. Which of the following statements is correct?
(1) Species diversity, in general, increases from poles to the equator.
(2) Conventional taxonomic methods are equally suitable for higher plants and microorganisms.
(3) India's share of global species diversity is about 18%.
(4) There are about 25,000 species of plants known in India.

151. Select the wrong statement.
(1) The number of species that are known and described range between 1.7-1.8 million.
(2) The scientific names ensure that each organism has only one name.
(3) The first word in a biological name represents the specific epithet while the second component denotes the genus.
(4) Biological names are generally in Latin and written in italics.

152. In nomenclature,
(1) both genus and species are printed in italics.
(2) genus and species are always of same name.
(3) both in genus and species, the first letter is capital.
(4) genus is written after the species.

153. The disadvantage of using common names for species is that
(1) the names may change.
(2) one name does not apply universally.
(3) one species may have several common names and one common name may be applied to two species.
(4) All of the above.

154. In referring to an organism in writing, such as in a newspaper, textbook or lab report, which of these rules should be followed?
(a) Underline or italicize genus
(b) Underline or italicize species
(c) First letter of species should be uppercase
(d) First letter of genus should be uppercase
(1) All except (c) (2) All except (d) (3) All except (a) (4) All except (b)

155. In trinomial nomenclature,
(1) The third term in animals represents the strain.
(2) The third term in plants denotes the variety.
(3) When species and sub-species are same, those names are called tautonyms.
(4) The third term in micro-organisms denotes the sub-species.

156. The third name in trinomial nomenclature in animals represents
(1) genus. (2) species. (3) sub-genus. (4) sub-species.

157. When the specific epithet exactly repeats generic name, it is called
(1) basinym. (2) synonym. (3) homonym. (4) tautonym.

158. Read the statements given below and identify the incorrect statement.
(1) Scientific names are used all over the world.
(2) Scientific names indicate relationship between species.

(3) Scientific names favour multiple naming for the same kind of an organism.
(4) Scientific names are often descriptive and tell us some important character of an organism.

159. The biological concept of species was given by
(1) John Ray. (2) Linnaeus. (3) Aristole. (4) Ernst Mayr.

160. ICBN is
(1) International Code of Biological Naming.
(2) International Code of Botanical Nomenclature.
(3) International Class of Biological Nomenclature.
(4) International Classification of Biological Nomenclature.

161. Scientific names of plants are based on principles and criteria agreed by and are given in
(1) IUCN. (2) ICZN. (3) ICBN. (4) ICPN

162. Identify the generic name from the given scientific names.
Panthera leo, Canis familiaris, Rosa indica
(1) Leo, Canis, indica (2) Panthera, Canis, Rosa
(3) Leo, familiaris, indica (4) Panthera, Rosa, familiaris

163. Which of the following is incorrect with respect to binomial nomenclature?
(1) Biological names are generally in Latin.
(2) The first word in a biological name represents the genus.
(3) Biological names are printed in italics.
(4) The first word of the genus starts with a small letter.

164. According to binomial nomenclature, every living organism has
(1) two scientific name with single component.
(2) one scientific name with two components.
(3) two names, one Latin and other common.
(4) one common name with three components.

165. Which one of the following statements given below is not included in universal rules of nomenclature?
(1) Generic names and specific epithet should be in Latin words.
(2) Generic name is immediately followed by name of taxonomists who described it first.
(3) Generic name must begin with capital letter.
(4) All letters of the specific name must be small.

166. Fill in the blanks by choosing the correct option
(a) Each may have one or more than one specific epithets representing different organisms.
(b) Taxonomic studies consider a group of individual organisms with fundamental similarities as a
(c) In case of plants, classes with a few similar characters are assigned to a higher category called
(d) being a higher category, is the assemblage of families which exhibit a few similar characters.

	(a)	(b)	(c)	(d)
(1)	species	order	phylum	Division
(2)	genus	species	division	Order
(3)	class	genus	species	Phylum
(4)	species	genus	division	Tribe

Select the mismatched pair.
(1) Order Primata

Gibbon, monkey, gorilla
(2) Order Carnivora
Tiger, cat, dog
(3) Order Polymoniales - Solanaceae and Liliaceae families
(4) Order Carnivora - Felidae and Canidae families

167. A taxon is a
(1) group of related families. (2) group of related species.
(3) type of living organisms. (4) taxonomic group of any ranking.

168. Which of the following statements about classification is not true?
(a) Members of a family are less similar than members of an included genus.
(b) An order has more members than the number of members in an included genus.
(c) Families have more members than phyla.
(d) Members of a family share a common ancestor in the more distant past than members of an included genus.
(e) The number of species in a taxon depends on their relative degree of similarity.
(1) Only (c) (2) Only (d) (3) Only (e) (4) None of these

169. Identify the correct sequence of taxonomic categories.
(1) Species - Order - Kingdom – Phylum (2) Species-Family-Genus - Class
(3) Genus - Species - Order – Phylum (4) Species - Genus - Order - Phylum

170. Two species which are morphologically almost identical but they do not interbreed are called
(1) Monotypic species. (2) Polytypic species.
(3) Evolutionary species. (4) Sibling species.

171. Biosystematics aims at
(1) the classification of organisms based on their evolutionary history and establishing their phylogeny studies.
(2) identification and arrangement of organisms on the basis of their cytological characteristics.
(3) the classification of organisms based on broad morphological characters.
(4) delimiting various taxa of organisms and establishing their relationships.

172. The practical purpose of classification of living organisms is to
(1) name the living organisms. (2) explain the origin of living organisms.
(3) trace the evolution of living organisms. (4) facilitate identification of unknown organisms.

173. When organism is in same division but is not in same order, the taxonomic term is called
(I) class. (2) genus, (3) species. (4) family,

174. Which of the following taxa covers a least number of organisms?
(I) Class (2) Order (3) Genus (4) Phylum

175. Linnacan system of classification was based on
(I) cytology. (2) morphology, (3) ecology. (4) embryology.

176. Sclentific study of diversity of organisms and their evolutionary relationships is called
(I) anatomy. (2) taxonomy. (3) systematics. (4) morphology.

177. Which of the following are not the processes that are basic to taxonomy?
(1) Characterisation (2) Systematics
(3) Identification (4) Nomenclature

178. Which one of the following has least similar characteristics?
(1) Family (2) Class (3) Genus (4) Species

179. Which one of the following is an odd category?
(1) Species (2) Class (3) Phylum (4) Sub family

180. The given table gives the classification of a wheat plant.

Kingdom	Plantae
Division	Angiospermae
(a)	Monocotyledonae
(b)	Poales
Family	(c)

181. Fill in the blanks using the correct option.
(1) (a) - Genus; (b) -Class; (c) -Poaceae
(2) (a) -Class; (b) -Order; (c) - Poaceae
(3) (a) - Genus; (b) - Class; (c) – Solanaceae
(4) (a) - Class; (b) - Order; (c) – Solanaceae

182. Who wrote Species Plantarium and provided a basis for classification of plants?
(1) Carolus Linnaeus
(2) Charles Darwin
(3) Robert Hooke
(4) Leeuwenhoek

183. Arrangement of taxa is termed as
(1) analogy.
(3) hierarchy.
(2) taxonomy.
(4) homology.

184. Each unit of a category of classification is termed as
(1) taxon.
(2) order,
(3) series.
(4) trophic level.

185. The correct sequence of taxa is:
(1) Class - Order - Family -Tribe - Genus - Species.
(2) Class - Order -Tribe - Family - Genus - Species.
(3) Phylum - Order - Class -Tribe - Genus - Species.
(4) Phylum -Tribe-Class - Order-Genus -Species.

186. The basic unit of taxonomy/biological classification is
(1) species.
(2) genus,
(3) phylum.
(4) order.

187. When organism is in the same class but is not in the same family, the taxonomic term is called
(1) order.
(2) genus.
(3) species.
(4) family.

188. Which of the following taxa covers a greater number of organisms?
(1) Class
(2) Order
(3) Genus
(4) Phylum

189. Animals are classified into hierarchical groups. In which one of the following, the largest number of species is found?
(1) Genus
(2) Order
(3) Family
(4) Phylum

190. Class is the category of taxonomy which includes related
(1) families.
(2) orders.
(3) genera.
(4) species.

191. In taxonomic hierarchy, which of the following group of taxa will have less number of similarities as compared to other?
(1) Solanaceae, Convolvulaceae and Poaceae
(2) Polymoniales, Poales and Sapindales
(3) Solanum, Petunia and Atropa
(4) Kéopard, tiger and lion

192. Match Column-I with Column-II and select the correct option.

Column-I	Column-II
(a) Homo sapiens	(i) Primata
(b) Mangifera indica	(ii) Poales

(c) Musca domestica	(iii) Sapindales
(d) Triticum aestivum	(iv) Diptera

(1) (a) - (i); (b) - (iii); (c) - (ii); (d) - (iv) (2) (a) - (iii); (b) - (ii); (c) - (iv); (d) - (i)
(3) (a) - (i); (b) - (iii); (c) - (iv); (d) - (ii) (4) (a) - (i); (b) - (iv); (c) - (iii); (d) - (ii)
Fill in the blanks (a) and (b) using the correct option.

193. Kingdom → Phylum → (a) → Order → (b)
(1) (a) -Genus; (b) -Species (2) (a) - Family; (b) - Class
(3) (a) - Class; (b) – Family (4) (a)-Species; (b) – Division

194. Taxonomy deals with
(1) the development of zoological parks.
(2) the study of kinds and diversity of microorganisms only.
(3) the evolutionary relationships between organisms.
(4) the classification of diverse organisms in different . taxa.

195. The correct sequence of taxonomic study of a newly discovered organism is
(1) first classification then identification, nomenclature and characterization.
(2) first identification then classification and then characterization and nomenclature.
(3) first nomenclature then characterization, identification and classification.
(4) first characterization then identification and nomenclature and then classification.

196. The equivalent rank of Carnivora in taxonomic categories of man and housefly is respectively
(1) Homo and Musca. (2) Hominidae and Muscidae.
(3) Mammalia and Insecta. (4) Primata and Diptera.

197. Which of the following suffix is used for category order?
(1) -phyta (2) -phyceae (3) -ales (4) -aceae

198. Place of keeping and studying dry plants is
(1) arboretum. (2) museum. (3) botanical garden. (4) herbarium.

199. An important function of botanical garden is
(1) providing beautiful area for recreation
(2) one can observe tropical plants over there
(3) they allow ex situ conservation of germplasm
(4) to allow reproduction freely with each other and form seeds

200. Which one of the following is not a correct statement?
(1) Herbarium houses are used to dry, press and preserve plant specimens.
(2) Botanical gardens have collection of living plants for reference.
(3) A museum has collection of photographs of plants and animals only.
(4) Key is a taxonomic aid for identification of specimens.

201. National Botanical Institute is located in
(I) Kolkata. (2) Mumbai. (3) Chennai. (4) Lucknow.

202. The Indian Botanical garden is located at
(1) Chennai. (2) Kolkata. (3) Lucknow. (4) Dehradun.

203. Which tazonomic aid gives comprehensive account of complete compiled information of a genus or family at a particular time?
(1) Flora (2) Monograph (3) Dictionary (4) Herbarium

204. Which one of the following is not a correct statement?
(1) Keys are analytical in nature.
(2) Zoological parks have collection of living animals for reference.
(3) A monograph gives a detailed account of all the plant species found in a particular geographical

area.
(4) Carolous Linnaeus is regarded as father of nomenclature.

205. Place of keeping and studying live plant species is
(l) aquarium. (2) museum. (3) botanical garden. (4) herbarium.

206. Lloyd Botanical Institute is located in
(1) Kolkata. (2) Mumbai. (3) Darjeeling. (4) Lucknow.

207. The Royal Botanical garden is located at
(1) Chile. (2) Kew. (3) UAE. (4) St. Louis.

208. Which one of the following is a taxonomical aid for identification of plants and animals based on similarities and dissimilarities?
(1) Flora (2) Keys (3) Manuals (4) Catalogues

209. Which of the following statement(s) about taxonomical aids is/are true?
(a) Keys are used to identify plants and animals based on similarities and dissimilarities.
(b) Flora contains the account of habitat and distribution of plants in a given area.
(c) Flora provide an index to the plant species found in a particular area.
(d) Monographs provide information for identifying the species found in an area.
(l) (a) and (b) (2) All except (d) (3) (a) and (d) (4) Only (a)

210. Herbarium sheets are arranged according to the system of classification and should have information about
(1) time and place of collection, English, local and botanical names, phylum, collector's name.
(2) date and time of collection, English, local and botanical names, class, collector's name.
(3) date and place of collection, English, local and botanical names, order, collector's name.
(4) date and place of collection, English, local and botanical names, family, collector's name.

211. The largest collection of herbarium in India is
(1) Blatter Herbarium, Mumbai. (2) Central Circle Herbarium, Allahabad.
(3) Central National Herbarium, Kolkata. (4) Southern Circle Herbarium, Coimbatore.

212. National Botanical Research Institute consists of
(1) dried and preserved plant specimens only.
(2) collection of preserved plant and animal specimens.
(3) flora, manuals and monographs only.
(4) collection of living plants for reference.

213. In museums, animals are
(1) kept and preserved in containers or jars. (2) preserved in boxes after killing.
(3) stuffed and then preserved. (4) All of the above.

214. Match Column-I with Column-II and choose the correct option.

Column-I	**Column-II**
(a) Ex-situ conservation	(i) Central National Herbarium
(b) Quick referral system	(ii) Museum
(c) Preserved plants and animals	(iii) Flora

(d) Actual account of habitat and distribution of plants in a given area	(iv) Royal Botanical Garden, Kew

(1) (a) - (ii); (b) - (iii); (c) - (iv); (d) - (i)
(2) (a)-(i); (b) - (iv); (c) - (ii); (d) - (iii)
(3) (a) - (iv); (b) - (i); (c) - (iii); (d) - (ii)
(4) (a) - (iv); (b) - (i); (c) - (ii); (d) - (iii)

215. Among the given taxonomic aids how many are associated with preservation of specimens?
Key, Flora, Museum, Botanical Garden, Catalogue, Herbarium, Zoological Park, Monograph
(1) One
(2) Three
(3) Two
(4) Four

216. Live specimens of plants and animals are found respectively in
(1) herbarium and museums.
(2) botanical garden and zoological parks.
(3) museums and herbarium.
(4) zoological parks and botanical gardens.

217. Key is
(1) a form of herbaria.
(2) a type of educational institute.
(3) a taxonomical aid used for identifying various organisms.
(4) a taxonomic category.

218. In zoological parks, animals are
(1) kept and preserved in containers or jars.
(2) preserved in boxes after killing.
(3) kept in protected environments under human care.
(4) stuffed and then preserved.

219. For identifying organisms through key usually
(1) two contrasting characters are used.
(2) one similar character is studied.
(3) two or more similar characters are used.
(4) only one statement called lead is used.

220. The largest Zoological Park in the world is located at
(1) S. America (Rio Jani).
(2) England (Cruger).
(3) Australia (Queen's Land).
(4) S. Africa (Victoria).

221. Dead specimens of plants and animals are found in
(1) herbarium and museums
(2) botanical garden and zoological parks
(3) museums and aquariums
(4) zoological parks and gene banks

222. Which of the following is not included under in situ conservation?
(1) National park
(2) Botanical garden
(3) Sanctuary
(4) Biosphere reserve

223. Match Column-I with Column-II and choose the correct option.

Column-I	**Column-II**
(a) Binomial nomenclature	(i) Carolus Linnaeus
(b) Generic name	(ii) Muscidae
(c) Family	(iii) Panthera
(d) Systema naturae	

(1) (a) - (i); (b) - (iii); (c) - (iii); (d) - (ii)
(2) (a) - (i); (b) - (iii); (c) - (ii); (d) - (i)
(3) (a) - (ii); (b) - (i); (c) - (i); (d) - (iii)
(4) (a) - (iii); (b) - (i); (c) - (ii); (d) - (i)

224. A prediction made by a scientist based on his observations is known as [Kerala PMT 2003]
(a) Law (b) Theory (c) Principle (d) Hypothesis

225. Melvin Calvin was professor of [AMU (Med.) 2012]
(a) Botany (b) Plant physiology (c) Chemistry (d) Blochemistry

226. The study of trends in human population growth and the prediction of future development is known as [Kerala PMT 2000]
(a) Sociology (b) Geography (c) Demography (d) Anthropology

227. Aquaculture does not include [BHU 2003]
(a) Pisces (b) Prawns (c) Silkworm (d) Shell fishery

228. Edaphology is [KCET 2007]
(a) Study of elephants (b) Study of snakes
(c) Study of amphibians (d) None of these

229. Animals that rely on the heat from the environment, rather than of metabolism, to raise their body temperature are, in the strict sense, called [AMU (Med.) 2010]
(a) Ectothermic (b) Poikilothermic
(c) Homeothermic (d) Endothermic

230. Study of ecology of population is called [MP PMT 2007]
(a) Autecology (b) Synecology (c) Ecotype (d) Demecology

231. Branch of Zoology dealing with the study of fishes is known as
[KCET 1994, 99; Bihar MDAT 1995; BCECE 1995;Odisha JEE 2002; MP PMT 2009]
(a) Herpetology (b) Ichthyology (c) Mammalogy (d) Ornithology

232. Science that deals with the study of external form, size, colour, structure and relative position of various parts of an organism is known as [Odisha JEE 2012]
[NCERT Pg. No.7]
(a) Ecology (b) Taxonomy (c) Anatomy (d) Morphology

233. In which type of isolation, two species living in different habitats are prevented from interbreeding
[J & K CET 2010]
(a) Temporal (b) Ecological (c) Behavioural (d) Gametic

234. The study of action of drug is known as [MP PMT 2001]
(a) Physiology (b) Pharmacology
(c) Phramacognosy (d) Pharmaceutical chemistry

235. Study of periodical changes in plants in relation to seasonal changes [DPMT 2006, 10]
(a) Physiognomy (b) Phycology (c) Phenology (d) Photoperiodism

236. Bloinformatics is an interdisciplinary branch which is concerned with the application of
[KCET 2006]
(a) Englneering techniques in biological studies
(b) Chemistry In understanding the biological phenomenon
(c) Physics in understanding various life processes
(d) Information science in analyzing the biological data

237. In medical science, the study of structure as well as various disorders of nervous system is termed as [Odisha JEE 2012]
(a) Neurology (b) Gynaecology (c) Cardiology (d) Endocrinology

238. The term "blology" was introduced by [MP PMT 1995]
(a) Aristotle (b) Darwin (c) Lamarck and Treviranus (d) Linnaeus

239. The scientist who cut the tail of mouse in many generations but found that it is not inherited
[CPMT 1995]
(a) Darwin (b) Bateson (c) Lamarck (d) Weismann

240. Who is called 'Father of Zoology' [MP PMT 1999]
(a) Aristotle (b) Darwin (c) Hippocrates (d) Theophrastus

241. Biometry refers to [AFMC 2012]
(a) Measurement of evolutionary rate in humans
(b) Measurement of living things and processes

(c) Measurement of fertility and mortality rate
(d) None of these

242. 'Father of Botany' is [Odisha JEE 2004, 11]
(a) Brunfels (b) Aristotle (c) Theophrastus (d) Linnaeus

243. K. Esau dominated in the field of plant biology up to the age of 99 years. She contributed mainly in the field of [AMU (Med.) 2012]
(a) Morphology of flowering plants (b) Anatomy of seed plants
(c) Classification of flowering plants (d) Physiology of seed plants

244. Study of behaviour of animals is called [RPMT 1996; BHU 2004]
(a) Ethology (b) Parapsychology (c) Euphenics (d) Etiology

245. William Harvey is known for the discovery of [HPMT 1993; Manipal 1996; AMU (Med.) 2000]
(a) Digestion (b) Respiration (c) Blood clotting (d) Blood circulation

246. The statement 'Nothing in biology makes sense except in the light of evolution' was given by [Kerala PMT 2000]
(a) A.I. Oparin (b) Th. Dobzhansky (c) Joseph Hooker (d) Charles Darwin

247. Study of fruit is known as [Odisha JEE 2010]
(a) Pomology (b) Palynology (c) Dendrology (d) Anthology

248. In history of biology, human genome project led to the development of [NCERT; CBSE PMT (Mains) 2011]
(a) Bloinformatics (b) Biosystematics (c) Biotechnology (d) Biomonitoring

249. Hybridoma technology was developed by IMP PMT 20011
(a) Taggart, 1982 (b) Vitella et al, 1982
(c) Prle and Saxton, 1987 (d) Milsteln and Kohler, 1975

250. Crick, one of the discoverer of DNA double hellcal atructure, was the man of [NCERT]
(a) Physics (b) Chemistry (c) Zoology (d) Botany

251. First experiment related to the method of hydroponics were done by (Keraln PMT 2000)
(a) Knop (b) HIII (c) Arnon (d) Sachs

252. Which branch study about remains of plant life [Odlsha JEE 1997; RPMT 2002; AFMC 2004]
(a) Palaentology (b) Palacobotany (c) Eugenics (d) Palynology

253. Name the sclentlst who was awarded the Nobel Prize for his genetic studies on the linear arrangement of genes on the chromosomes in the frultny, Drosophilla melanogaster [Kerala PMT 2001]
(a) C.F. Wolff (b) T.A. Knight (c) J. Swammerdam (d) T. H. Morgan

254. Who wrote the book 'Genetics and Origin of Specles' [MP PMT 1999; Pb. PMT 1999]
(a) R.A. Fisher (b) G.L. StebbIns (c) Th. Dobzhaansky (d) J.B.S. Haldane

255. Study of Ticks and Mites is called (MP PMT 2002; Odisha JEE 2005)
(a) Acarology (b) Entomology (c) Malacology (d) Carcinology

256. Which one of the following aspects is an exclusive characteristlc of llving things [CBSE PMT (Malns) 2011] [NCERT Pg. No.3]
(a) Perception of events happening in the environment and their memory
(b) Increase in mass by accumulation of materlal both on surface as well as Internally
(c) Isolated metabolic reactions occur in viltro
(d) Increase in mass from Inside only

257. Study the four statements $(A-D)$ glven below and select the two correct ones out of them
(A) Definition of blological species was glven by Ernst Mayr
(B) Photoperlod does not affect reproduction In plants
(C) Binomial nomenclature system was glven by R. H. Whittaker
(D) In unicellular organisms, reproduction is synonymous with growth
The two correct statements are [NEET (Phase-II) 2016] [NCERT Pg. No.6]
(a) (A) and (B) (b) (B) and (C) (c) (C) and (D) (d) (A) and (D)

258. When spontaneous process occurs then free energy of a system [DPMT 2003]
(a) Decreases (b) Increases
(c) Remains same (d) Elther can decreases or Increase

259. A molecule is reduced means it (Odisha JEE 2008, 0%
(a) Loses electron (b) Galns proton
(c) Loses proton and electrons (d) Gains electron

260. The type of bond Involved in the formation of sodium chloride is (Keraln PMT 2000)
(a) Ester bond (b) Peptide bond
(c) lonic bond (d) Covalent bond
(c) Hydrogen bond

261. Animals spending winter in dormant conditions is referred is under
(J & K CET 2005; RPMT 2005)
(a) Hibernation (b) Aestivation (c) Mimicry (d) Camouflage

262. The type of linkage present in carbohydrates is [CPMT 2005]
(a) Peptide (b) Glycosidic (c) Amide (d) Phosphate bonds

263. In ATP high energy bond is present (AMU (Med.) 2001; KCET 2002; CPMT 2005)
(a) Between nucleoside and phosphate group
(b) Between sugar and phosphate group
(c) Between base (Adenine) and phosphate group
(d) None of these

264. Anabolism is [BHU 2000]
(a) Endergonic process (b) Exergonic process
(c) Bidirectional process (d) Destructive process

265. Which of the following is a trace element required only in small amounts by most living things
(a) Oxygen (b) Iron (c) Nitrogen (d) Hydrogen

266. The bond formed between the first phosphate group and adenosine in ATP is [KCET 2001]
(a) Phosphoester bond (b) Adenophosphate bond
(c) Nitrophosphate bond (d) Phosphoanhydride bond

267. Carbohydrates, the most abundant blomolecules on earth, are produced by [CBSE PMT 2005]
(a) Some bacteria, algae and green plant cells (b) Fungl, algae and green plant cells
(c) All bacteria, fungi and algae (d) Viruses, fungi and bacteria

268. Which of the following chemical characteristics is not common to all living beings [Manipal 2003]
(a) Type of protein present in the body (b) Similar triple code for amino acids
(c) Energy is stored by high phosphate bonds (d) Ribosomes are the sites of protein synthesis

269. Some plants having pleasant odour and attractive colours for [HPMT 2000]
(a) Hydrophily (b) Anemophily (c) Entomophily (d) None of these

270. Metabolism comprises [NCERT Pg. No.4]
(a) Digestion of food (b) Elimination of wastes
(c) Exchange of gases (d) Various energy exchanges in cell

271. Biological organisation starts with
[CBSE PMT 2007] [NCERT Pg. No.5]
(a) Submicroscopic molecular level (b) Cellular level
(c) Organismic level (d) Atomic level

272. The total heat content of a system is [KCET 2002]
(a) Entropy (b) Free energy (c) Enthalpy (d) Kinetic energy

273. Which of the following is the main adaptation for a plant to survive in xerophytic condition [HPMT 2000]
(a) Spines (b) No stomata (c) Stipular leaves (d) None of the above

274. The energy transformation in the nervous system is [Kerala PMT 2002]

(a) Chemical to radiant
(b) Chemical to electrical
(c) Chemical to mechanical
(d) Mechanical to radiant

275. Maintenance of internal favourable conditions, despite changes in external environment is [AFMC 1999; AMU (Med.) 2000; DPMT 2001; Kerala PMT 2001, 03]
(a) Entropy (b) Steady state (c) Enthalpy (d) Homeostasis

276. A coconut high in a palm tree has........owing to its location
(a) Free energy (b) Kinetic energy (c) Activation energy (d) Potential energy

277. Which statement is correct for biomolecules [RPMT 2001]
(a) DNA is a polymer of ribonucleotides
(b) All carbohydrates break down into glucose
(c) RNA is single stranded and contain different purine base than DNA
(d) Sequence of amino acids determine primary structure of proteins

278. Reproduction of immature or larval stage of animals caused by the acceleration of maturation is called
(a) Cladogenesis (b) Paedogenesis (c) Morphogenesis (d) Parthenogenesis

279. Cause of mimicry is [CBSE PMT 2002]
(a) Isolation (b) Attack (c) Protection (d) Both (b) and (c)

280. The basis of life (secret of life) is [AFMC 2002]
(a) Lipid (b) Protein (c) Nucleic acid (d) Nucleoprotein

281. Lipids are generally
(a) A high energy source
(b) Hydrophobic
(c) Composed of fatty acids
(d) All of the above

282. Glycogen is most structurally similar to
(a) Glucose (b) Starch (c) Maltose (d) Cellulose

283. Which of the following is produced only by plants
(a) Starch (b) Glycogen (c) Cholesterol (d) Triglycerides

284. The living organisms can be unexceptionally distinguished from the non-living things on the basis of their ability for [NCERT Pg. No.4] [CBSE PMT 2007]
(a) Responsiveness to touch
(b) Interaction with the environment and progressive evolution
(c) Reproduction
(d) Growth and movement

285. Which one of the following sequences is true [Pb. PMT 2000; VVMC Safdarjung 2001]
(a) Observations, problem defining, hypothesis, experiment
(b) Experiment, hypothesis, problem defining, observation
(c) Observation hypothesis problem defining, experiment
(d) Problem defining, observation, hypothesis, experiment

286. The process in which excess energy is lost by light waves is called [JIPMER 2002]
(a) Photolysis (b) Fluorescence (c) Photo-oxidation (d) Photophosphorylation

287. Among the energy values or nutrients, 9.3 calories is that of [JIPMER 2002]
(a) Fats (b) Proteins (c) Vitamins (d) Carbohydrates

288. The label of a herbarium sheet does not carry information on [NEET (Phase-II) 2016] [NCERT Pg. No.11]
(a) Height of the plant
(b) Date of collection
(c) Name of collector
(d) Local names

289. The branch of Botany concerned with the classification, nomenclature and identification of plants is [NCERT Pg. No.8]
(a) Systematic Botany (b) Ecology (c) Morphology (d) Physiology

290. Which one of the following is not a correct statements [NCERT Pg. No.12,13] [NEET 2013]
(a) Key is taxonomic aid for identification of specimens

(b) Herbarium houses dried, pressed and preserved plant specimens
(c) Botanical gardens have collection of living plants for reference
(d) A museum has collection of photographs of plants and animals

291. A person who studies about the origin, evolution and variations in plants and also about the classification of plants, is called as [NCERT Pg. No.6] [AIIMS 1992]
(a) Classical taxonomist (b) Herbal taxonomist
(c) α-taxonomist (d) β-taxonomist

292. ICBN stands for
[NCERT Pg. No.6] [MP PMT 2003; BVP 2004; CBSE PMT 2007]
(a) International Council for Botanical Nature
(b) International Code for Botanical Nomenclature
(c) Indian Code of Botanical Nomenclature
(d) None of the above

293. Who amongst the following is regarded as the "Father of Taxonomy"
(DPMT 1992; MP PMT 1997; HPMT 1998; WB JEE 2012) [NCERT Pg. No.6]
(a) Takhtajan (b) Linnaeus (c) Bentham and Hooker (d) Theophrastus

294. If a botanist has to nomenclate a similar species, he will use (CPMT 1999; JIPMER 2001)
(a) Syntype (b) Neotype (c) Mesotype (d) Isotype

295. Which of the following taxonomical ranks contain organisms least similar to one another
[DPMT 1999; Pb. PMT 2000] [NCERT Pg. No.10]
(a) Class (b) Genus (c) Family (d) Species

296. In a hierarchical system of plant classification, which one of the following taxonomic ranks generally ends in 'ceae'[AFMC 2003] [NCERT Pg. No.11]
(a) Family (b) Genus (c) Order (d) Class

297. Which one of the following is a taxonomical aid for identification of plants and animals based on similarities and dissimilarities [NCERT Pg. No.13] [Kerala PMT 2012]
(a) Flora (b) Keys (c) Monographs (d) Catalogues
(e) Manuals

298. When organism is in same class but is not in same family, the taxonomic term is called as
[Odisha JEE 2008] [NCERT Pg. No.10]
(a) Order (b) Genus (c) Species (d) Family

299. Which is a taxon
[CBSE PMT 1992; Pb. PMT 1998] [NCERT Pg. No.8]
(a) Genera (b) Family (c) Class (d) None of these

300. A group of plants with similar traits of any rank is
[CBSE PMT 1990, 92, 96, 97; AFMC 1994;
CPMT 1996; Chd. CET 2000; Pb. PMT 2002; DPMT 2006; Odisha JEE 2009]

(a) Species (b) Genus (c) Order (d) Taxon

301. Animals are classified into hierarchical group, in which one of the following the largest number of species is found [WB JEE 2012]
(a) Genus (b) Order (c) Family (d) Cohort

302. In Botanical nomenclature of plants
[MP PMT 1993] [NCERT Pg. No.7] .
(a) Genus is written after the species
(b) Both in genus and species the first letter is a capital letter
(c) Genus and species may be same name
(d) Both genus and species are printed in italics

303. Match the following
(A) Genera Plantarum (1) Aristotle
(B) Specles Plantarum (2) LInnaeus
(C) Historla Generalis (3) Bentham and Hooker Plantarum
(D) Scala Naturae (4) Pliny
(5) John Ray
[CBSE PMT 1999, 2001; AIEEE Pharmacy 2004; Odisha JEE 2004; WB JEE 2008: VITEEE 2008]

(a) A-4, B-2, C-5, D-3
(b) A-4, B-2, C-1, D-3
(c) A-4, B-2, C-3, D-1
(d) A-3, B-2, C-5, D-1

304. The smallest taxon is called
[NCERT Pg. No.9] [J & K CET 2008; MP PMT 2010, 11]
(a) Class (b) Order (c) Genus (d) Species

305. A system of classification in which a large number of traits are considered is
[AIIMS 1996; CBSE PMT 1999] [NCERT Pg. No.6]
(a) Artificial system
(b) Synthetic system
(c) Natural system
(d) Phylogenetic system

306. In Bentham and Hooker's classification what was positioned in between dicots and monocots
[AFMC 2008]
(a) Gymnosperms
(b) Bryophytes
(c) Algae
(d) Pteriodophytes

307. Who proposed the Binomial Nomenclature System
[RPMT 1992, 98; CBSE PMT 1993, 94; AMU (Med.) 1995, 98, 2000; Pb. PMT 1998, 99, 2000; CPMT 1999, 2000; KCET 1999, 2000; J & K CET 2000; Wardha 2000; AIIMS 2001; MHCET 2001, 02,04;
BVP 2002; MP PMT 2003, 06; Odisha JEE 2004, 11;
DPMT 2004; BHU 2005, 06; AFMC 2010]
[NCERT Pg. No.6]
(a) Whittaker (b) Mendel (c) Carl Linnaeus (d) Tippo

308. New Systematics introduced by Sir Julian Huxley is also called [Kerala PMT 2008]
(a) Phenetics
(b) Cladistics
(c) Biosystematics
(d) Numerical taxonomy
(e) Chemotaxonomy

309. Binomial nomenclature system of Linnaeus means that every organism has
[CBSE PMT 1993, 94;
BHU 1994, 2002; RPMT 1995; CPMT 1995; APMEE 1995; EAMCET 1995; AMU (Med.) 1995; DPMT 1996; Chd. CET 2000; Pb PMT 2000; MP PMT 2004, 06, 12; PET (Pharmacy) 2013]
[NCERT Pg. No.6]
(a) One name given by two scientists
(b) Two names one Latin and other of a person
(c) Two names one scientific and other popular
(d) One scientific name with generic and other with specific epithet

310. Which of the following is a merit in the Bentham and Hooker's system of classification
(Kerala PMT 2008)
(a) The position of Gymnospermae in between dicots and monocots
(b) Closely related families are placed apart
(c) The placement of family asteraceae in the beginning of gamopetalae
(d) The placement of order ranales in the beginning
(e) The placement of orchidaceae in microspermae

311. Linnaeus system of plant classification is [MP PMT 1996, 97; AIIMS 1999; Kerala PMT 2004]
[NCERT Pg. No.6]
(a) Artificial (b) Natural (c) Phylogenetic (d) None of the above

312. In plants Latin names are suggested because [NCERT Pg. No.7]
(a) Latin is a simple language
(b) In whole world there would be only one name for one plant
(c) It is easy to write
(d) Most of the names in other languages are not correct

313. Specific epithet is [NCERT Pg. No.6] [Odisha JEE 2012]
(a) First word in the scientific name of a species
(b) Second word in the scientific name of a species
(c) Both (a) and (b)
(d) None of these

314. Who did propose natural classifications of plants
[CPMT 1992, 2009; MHCET 2002; Odisha JEE 2004, 09; BVP 2004; KCET 2012]

[NCERT Pg. No.6]
(a) Carolus Linnaeus (b) John Hutchinson
(c) Bentham and Hooker (d) Oswald Tippo

315. Who proposed phylogenetic classification of plants
[BHU 1991; CPMT 1998;
Haryana PMT 2000; Pb. PMT 2004] [NCERT Pg. No.18]
(a) Linnaeus (b) Hutchinson (c) Bentham and Hooker (d) Mehta

316. Identify from the following, the only taxonomic category that has a real existence.
[NCERT Pg. No.9] [KCET 2006]
(a) Genus (b) Species (c) Phylum (d) Kingdom

317. Five kingdom classification was proposed by
[Haryana PMT 1993; AFMC 1994; MP PMT 2000,06; MHCET 2001; BVP 2001;
Kerala PMT 2004; Odisha JEE 2004;
J & K CET 2010; CPMT 2010] [NCERT Pg. No.17]
(a) Birbal Sahni (b) Whittaker (c) Aristotle (d) Oswald Tippo

318. In Whittaker's 'Five Kingdom Classification', eukaryotes were assigned to
[BHU 1994; KCET 1999; MHCET 2003; Odisha JEE 2005] [NCERT Pg. No.17]
(a) Only two of the five kingdoms (b) Only three of the five kingdoms
(c) Only four of the five kingdoms (d) All the five kingdoms

319. In five kingdom system, the main basis of classification is
[Manipal 2001; CBSE PMT 2002] [NCERT Pg. No.17]
(a) Nutrition (b) Nucleus structure
(c) Cell wall structure (d) Asexual reproduction

320. Brinjal, potato, tomato, onion, ginger belongs to
[NCERT P.No. 9] [Odisha JEE 2009]
(a) Single family (b) Species (c) Genera (d) Same genus

321. Two kingdoms constantly figured in all biological classifications are
[NCERT Pg. No.16] [J & K CET 2008]
(a) Plantae and animalia (b) Monera and animalia
(c) Protista and animalia (d) Protista and plantae

322. Which of the following is grouped under phanerogams [CBSE PMT 2000]
(a) Pteridophytes (b) Gymnosperms (c) Angiosperms (d) Both (b) and (c)

323. Single-celled eukaryotes are included in [NCERT Pg. No.20]
[DUMET 2009; CBSE PMT (Pre.) 2010; BHU 2012]
(a) Monera (b) Protista (c) Fungi (d) Archaea

324. No. of plants species reported in India are [DPMT 2003]
(a) 45,000 (b) 40,000 (c) 90,000, (d) 50,000

325. Genus is a group of similar and related
[Pb. PMT 2000] [NCERT Pg. No.9]
(a) Order (b) Genera (c) Families (d) Species

326. Which of the following is excluded in Whittaker's five kingdom system of classification
[BHU 2012] [NCERT Pg. No.17]
(a) Viruses (b) Algae (c) Fungi (d) Bacteria

327. Systema Naturae is concerned with
[MP PMT 1998] [NCERT Pg. No.8]
(a) Solar system (b) Ecosystem
(c) Classification of plants and animals (d) Natural selection

328. Which of the following are correctly matched with respect to their taxonomic classification
[NEET 2013]
(a) Spiny anteater, sea urchin, sea cucumber Echinodermata
(b) Flying fish, cuttlefish, silverfish - Pisces
(c) Centipede, millipede, spider, scorpion - Insecta
(d) Housefly, butterfly, tsetsefly, silverfish – Insecta

329. The meaning of taxon in the classification of animals [NCERT Pg. No.8] [CBSE PMT 1996]
(a) A group of same species
(b) A group of animals on the basis of number of chromosomes
(c) A group of same type of animals
(d) A group of similar genera

330. Most important criteria used for the present day classification of living organisms is based on [CBSE PMT 1991]
(a) Presence and absence of notochord (b) Resemblances in external features
(c) Breeding habits (d) Anatomical and physiological characteristics

331. The non-nucleated, unicellular organisms of Whittaker's (1969) classification are included in the kingdom
[MP PMT 1994; BHU 1997, 2012;
Kerala PMT 2000: J & K CET 2008: AMU (Med.) 2010] [NCERT Pg. No. 18]
(a) Protista (b) Monera (c) Animalia (d) Plantae

332. The suffix 'idae' refers to [NCERT Pg. No.11]
(a) Family (b) Genus (c) Order (d) Division

333. Read the statements given below and identify the incorrect statement
[NCERT Pg. No.7] [KCET 2012]
(a) Scientific names are used all over the world
(b) Scientific names are often descriptive and tell us some important character of an organism
(c) Scientific names indicate relationship between species
(d) Scientific names favour multiple naming for the same kind of an organism

334. The term 'New systematics' was introduced in 1940 by[AFMC 1993; AMU (Med.) 1999; BVP 2002]
(a) Adolf Engler (b) Karl prantl (c) George Bentham (d) Julian Huxley

335. Taxonomic hierarchy refers to
[DUMET 2009] [NCERT Pg. No.8]
(a) Stepwise arrangement of all categories for classification of plants and animals
(b) A group of senior taxonomists who decide the nomenclature of plants and animals
(c) A list of botanists or zoologists who have worked on taxonomy of a species or group
(d) Classification of a species based on fossil record

336. The third name of the trinomial nomenclature is of [NCERT; JIPMER 1993; APMEE 2001]
(a) Sub-genus (b) Species (c) Sub-species (d) Type

337. In which book has "binomial nomenclature" been used for the first time
[NCERT Pg. No.6, 8] [MP PMT 1999; Pb PMT 1999; Odisha JEE 2005]
(a) Histoire naturelle (b) Systema naturae
(c) Historia naturalis (d) Historia plantarum

338. Who developed the "key" for identification of animals [MP PMT 1999]
(a) John Ray (b) Goethe (c) Georges Cuvier (d) Theophrastus

339. What is the name of the book written by Aristotle [MP PMT 1999; Pb PMT 1999]
(a) Historia Animalium (b) Histoire Naturelle
(c) Systema Naturae (d) Philosophic Zoologique

340. The replacement of two kingdoms grouping by five kingdom classification was proposed in the year [NCERT Pg. No.10, 17] [Kerala CET 2003]
(a) 1859 (b) 1758 (c) 1919 (d) 1969

341. Engler and Prantl published a phylogenetic system in monograph [Kerala CET 2005]
(a) Die Naturlichen Pflanzen Famllien (b) Historia Plantarum
(c) Species Plantarum (d) Genera Plantarum
(e) Origin of Species

342. In the flue-kingdom classification, Chlamydomonas and Chlorella have been Included In [CBSE PMT (Mains) 2012] [NCERT Pg. No.18]
(a) Protista (b) Algae (c) Plantae (d) Monera

343. Which structure is present in both prokaryotic and eukaryotic plant cells
[NCERT Pg. No.17] [WB JEE 2008]
(a) Cell wall (b) Nucleus (c) Chloroplast (d) Mitochondria

344. Oryza sativa is the binomial name of the rice plant, the sativa stands for
[NCERT Pg. No.9] [WB JEE 2008]
(a) Specific name (b) Specific epithet (c) Species name (d) Specific nomenclature

345. Which of the following is considered as neither prokaryotes nor eukaryotes
[NCERT Pg. No.26] [J & K CET 2010]
(a) Bacteriophages (b) Bacteria (c) Monera (d) Fungi

346. Interbreeding natural population of animals are referred to as belonging to the same
[NCERT Pg. No.9] [AMU (Med.) 2002; J & K CET 2010]
(a) Family (b) Species (c) Genus (d) Variety

347. Natural system of classification is based on [DPMT 1993] [NCERT Pg. No.19]
(a) Morphology (b) Phylogeny
(c) Morphology and affinities (d) Ontogeny
348. Which one of the following organisms is scientifically correctly named, correctly printed according to the International Rules of Nomenclature and correctly described
[NCERT Pg. No.22] [CBSE PMT (Mains) 2012]
(a) Musca domestica - The common house lizard, a reptile
(b) Plasmodium falciparum - A protozoan pathogen causing the most serious type of malaria
(c) Felis tigris - The Indian tiger, well protected in Gir forests
(d) E.coli - Full name Entamoeba coli, a commonly occurring bacterium in human intestine
349. Consider the following statements with respect to characteristic features of the kingdom
A. In animalia the mode of nutrition is autotrophic
B. In monera the nuclear membrane is present
C. In protista the cell type is prokaryotic
D. In plantae the cell wall is present
Of the above statements [NCERT P.No.17] [Kerala PMT 2012]
(a) A alone is correct (b) B alone is correct
(c) C alone is correct (d) D alone is correct
(e) A, B and C are correct
350. Cohort is a group correlated with [MHCET 2004]
(a) Species (b) Genera (c) Familles (d) Order
351. International applicable to
(a) Plants (b) Animals
(c) Both animals and plants (d) None of the above
NCERT Pg. No.6, 7]
lature" is code of "Biological
352. The term "phylum" in animal classification was coined by
[CBSE PMT 1992; MP PMT 1994, 2010] [NCERT Pg. No.10]
(a) E. Haeckel (b) John Ray
(c) G.L. Cuvier (d) Carolus Linnaeus
353. Family is placed between
[Odisha JEE 2011] [NCERT Pg. No.10]
(a) Order and genus (b) Genus and species
(c) Class and order (d) Phylum and class
354. The common characteristics between tomato and potato will be maximum at the level of their
(a) Family (b) Order (c) Division (d) Genus
[NEET (Karnataka) 2013]
355. Five kingdom system of classification suggested by R.H. Whittaker is not based on
[CBSE PMT 2014] [NCERT Pg. No.17]
(a) Mode of nutrition (b) Complexity of body organization
(c) Presence or absence of a well defined nucleus (d) Mode of reproduction
356. Cytotaxonomy is connected with [NCERT Pg. No.30]
(a) Chemical composition of cytoplasm (b) Cell organelles
(c) Cytochromes (d) Shape and size of cells
357. 'Taxa' differs from 'taxon' due to
[DUMET 2010] [NCERT Pg. No.8]
(a) This being a higher taxonomic category than taxon
(b) This being lower taxonomic category than taxon
(c) This being the plural of taxon
(d) This being the singular of taxon
358. The term 'biosystematics' was coined by [J & K CET 2010; MP PMT 2010, 12]
(a) Gaspard Bauhin (b) Camp and Gilly
(c) Karl Prantl (d) Robert Brown
359. Which of the following statements regarding universal rules of nomenclature is wrong
[NCERT Pg. No.7] [Kerala PMT 2010; NEET (Phase-J) 2016]
(a) The first word in a biological name represents the genus
(b) The first word denoting the genus starts with a capital letter
(c) Both the words in a biological name, when handwritten, are separately underlined
(d) Biological names are generally in Greek and can be written in any language
(e) The second component in a biological name denotes the specific epithet

360. Phenetic classification of organisms is based on
[CBSE PMT 2004; WB JEE 2016] [NCERT Pg. No.7]
(a) Dendogram based on DNA characteristics(b) Sexual characteristics
(c) Observable characteristics of existing organisms
(d) The ancestral lineage of existing organisms

361. Classical systematics embodies/concept of classical taxonomist is
(a) Biological concepts (b) Species concept
(c) Typological concept (d) All the above
[NCERT Pg. No.8] [JIPMER 1997]

362. Species is
[NCERT Pg. No.9] [CBSE PMT 1994]
(a) Specific unit of evolution
(b) Specific unit in the evolutionary history of a race
(c) Specific class of evolution
(d) Not related to evolution

363. Two plants are taxonomically related if [NCERT]
(a) They store carbohydrate in the same type of molecule
(b) Both obtain energy from hydrolysis of ATP into ADP and inorganic phosphate
(c) Both have similarly lobed palmate leaves
(d) Both have pinnately veined leaves

364. A group of related genera, with still less number of similarities as compared to the genus and species, constitutes
[AFMC 2009; DUMET 2010] [NCERT Pg. No.9]
(a) Order (b) Class (c) Family (d) Division

365. Match the following and choose the correct combination from the options given

Column I (Common name)		**Column II (Taxonomic category Order)**	
A.	Wheat	1.	Primata
B.	Mango	2.	Diptera
C.	Housefly	3.	Sapindales
D.	Man	4.	Poales

[NCERT Pg. No.11] [Kerala PMT 2010]
(a) A-1, B-2, C-4, D-3 (b) A-4, B-3, C-2, D-1
(c) A-2, B-4, C-1, D-3 (d) A-3, B-4, C-2, D-1
(e) A-4, B-2, C-3, D-1

366. The term species was coined by [BVP 2002; KCET 2004]
(a) Aristotle (b) Engler (c) John Ray (d) Linnaeus

367. Plant classification proposed by Carolus Linnaeus was artificial because it was based on [RPMT 1990; CPMT 1995] [NCERT Pg. No.16]
(a) Only a few morphological characters
(b) Evolutionary tendencies which are diverse
(c) Anatomical characters which are adaptive in nature
(d) Physiological traits alongwith morphological characters

368. Classification of organisms based on evolutionary as well as genetic relationships is called
[DUMET 2010]
(a) Biosystematics (b) Phenetics (c) Numerical taxonomy (d) Cladistics

369. Match the following and select the correct combination from the options given below

Column I (Kingdom)		Column II (Class)	
A.	Plantae	1.	Archaebacteria

B.	Fungi	2.	Euglenoids
C.	Protista	3.	Phycomycetes
D.	Monera	4.	Algae

[NCERT Pg. No.17][Kerala PMT 2011]
(a) A-4, B-3, C-2, D-1
(b) A-1, B-2, C-3, D-4
(c) A-3, B-4, C-2, D-1
(d) A-4, B-2, C-3, D-1
(e) A-2, B-3, C-4, D-1

370. First act in taxonomy is [NCERT Pg. No.8] [Wardha 2002]
(a) Description (b) Identification (c) Naming (d) Classification
371. Taxonomy based on determination of genetic relationships is
[NCERT Pg. No.8] [JIPMER 1997]
(a) Cytotaxonomy
(b) Numerical taxonomy
(c) Biochemical taxonomy
(d) Experimental taxonomy
372. Branch connected with nomenclature, identification and classification is [NCERT Pg. No.8]
[CPMT 1991; AMU (Med.) 2000; Kerala PMT 2002] Or
The study of theory, practice and rules of classification of living and extinct organisms is called [Odisha JEE 2012]
(a) Ecology (b) Taxonomy (c) Morphology (d) Physiology
373. Sequence of taxonomic categories is [NCERT Pg. No.8]
[DPMT 1992; CBSE PMT 1992;
AFMC 1992, 2001; HPMT 1994; Pb. PMT 1997; KCET 2001, 11, 18; Kerala PMT 2007; AMU (Med.) 2012]
(a) Class - phylum - tribe - order - family - genus - species
(b) Division - class - family - tribe - order - genus - species
(c) Division - class - order - family - tribe - genus - species
(d) Phylum - order - class - tribe - family - genus – species
374. Phylogenetic system of classification is based on
[CBSE PMT 1994, 2006, 09; Pb. PMT 2000;
Odisha JEE 2002; DPMT 2006] [NCERT Pg. No.8, 30]
(a) Evolutionary relationships
(b) Morphological features
(c) Chemical constituents
(d) Floral characters
375. An attribute found in plants but not animals is
(a) Metabolism
(b) Sexual reproduction
(c) Autotrophy
(d) Asexual reproduction
376. Systema Naturae was written by
[CPMT 1991, 92, 93;
Kerala PMT 2010; Odisha JEE 2010] [NCERT Pg. No.8]
(a) Lamarck (b) Cuvier (c) Aristotle (d) Linnaeus
377. Which of the following taxonomist described classification of plant kingdom in "Families of flowering plants" [CPMT 2004]
(a) Cronquist (b) Takhtajan (c) Benson (d) Hutchinson
378. Mayr's biological concepts of species is mainly based on
[NCERT; BVP 2004; BHU 2004, 08; CPMT 2009]
(a) Morphological traits
(b) Reproductive isolation
(c) Modes of reproduction
(d) Morphology and reproduction
379. Two morophologically similar populations are intersterile. They belong to
[NCERT; BHU 1994, 97, 2000; Kerala PMT 2005)
(a) One species
(b) Two biospecies
(c) Two sibling species
(d) None of the above
380. Which one possess characters of both plants and animals [CBSE PMT 1995]
381. A unicellular organism often considered connecting link between plants and animals is [AFMC 1997; JIPMER 1998]
(a) Bacteria (b) Mycoplasma (c) Paramecium (d) Euglena
382. Distinction of prokaryota and eukaryota is mainly based on
[NCERT Pg. No.17,18] [MP PMT 1995, 98]
(a) Nucleus only
(b) Cell organelles only
(c) Chromosomes only
(d) All the above

383. Algae with photosynthetic pigments possess nutrition
[AMU (Med.) 1997; Manipal 1997] [NCERT Pg. No.32]
(a) Holozoic (b) Saprophytic (c) Holophytic (d) Parasitic
384. Select the incorrect statements
(A) Lower the taxon, more are the characteristics that the members within the taxon share
(B) Order is the assemblage of genera which exhibit a few similar characters
(C) Cat and dog are included in the same family Felidae
(D) Binomial nomenclature was introduced by Carolus Linnaeus
[NCERT Pg. No.9][Kerala PMT 2011]
(a) A, B and C (b) B, C and D (c) A and D (d) C and D
(e) B and C
385. Phylogenetic system was given [DPMT 2007; Kerala PMT 2009; CPMT 2010]
(a) Engler & Prantl (b) Pliny (c) John Ray (d) R.H. Whittaker
386. Huxley is father of [MP PMT 2007]
(a) Classical taxonomy (b) Artificial taxonomy
(c) Neo-taxonomy (d) Adansonian taxonomy
387. Species can be identified on the basis of
[MP PMT 2007] [NCERT Pg. No.9]
(a) Interbreed (b) Species diversity
(c) Reproductive isolation (d) None of these
388. In the five kingdom system of classification, which single kingdom out of the following can include blue-green algae, nitrogen fixing bacteria and methanogenic archaebacteria
[NCERT Pg. No.17][CBSE PMT 1998; Pb. PMT 1998]
(a) Monera (b) Fungi (c) Plantae (d) Protista
389. Floral features are commonly used for identification of angiosperms because
[NCERT; CBSE PMT 1998]
(a) Reproductive parts are more conservative
(b) Flowers can be safely pressed
(c) Flowers are nice to work with
(d) Flowers have various colours and scents
390. Two similar holotypes are called [BHU 1997]
(a) Mesotypes (b) Meotypes (c) Syntypes (d) Isotypes
391. Binomial system of nomenclature for plants is effective from [JIPMER 1997]
(a) 5.8.1771 (b) 1.5 .1753 (c) 1.8 .1758 (d) 6.7.1736
392. A system of classification that is based on evolution, order and ancestry is known as
[Pb. PMT 1998; MP PMT 1998] [NCERT Pg. No.30]
(a) Natural system (b) Analogous system
(c) Phylogenetic system (d) Homologous system
393. Characteristics which delimit a family are more general than those which delimit a[JIPMER 1999]
(a) Cohort (b) Phylum (c) Class (d) Genus
394. First great taxonomist was [NCERT Pg. No.8] [BHU 1999]
(a) Linnaeus (b) Hooker (c) Aristotle (d) Engler
395. In the classification of plants, the term cladistics refers to the [Kerala PMT 2006; BHU 2008]
(a) Phylogenetic classification (b) Sexual classification
(c) Artificial classification (d) Natural classification
(e) Binomial classification
396. Five kingdom classification includes
[DPMT 2006] [NCERT Pg. No.17]
(a) Monera, Protista, Fungi, Plantae Animalia
(b) Algae, Fungi, Bryophyta, Pteridophyta, Gymnosperms
(c) Virus, Prokaryota, Fungi, Plantae, Animalia
(d) Monera, Protista, Animalia, Plantae, Alage
397. Which of the following is less general in characters as compared to genus
[CBSE PMT 2001] [NCERT Pg. No.9]
(a) Species (b) Family (c) Class (d) Division
398. What is correct
[NCERT Pg. No.9] [Manipal 2001; Odisha JEE 2012; KCET 2012]
(a) Apis indica (b) trypanosoma gambiense
(c) Ficus Bengalensis (d) Mangifera indica
399. Which covers the largest number of organisms [NCERT Pg. No.10] [Kerala PMT 2001]
(a) Genus (b) Family (c) Phylum (d) Class (e) Order

400. Which of the following is not taxon but a category [NCERT Pg. No.10] [MHCET 2000]
(a) Division (b) Angiosperms (c) Polypetalae (d) Hibiscus

401. Chemotaxonomy is connected with [NCERT; MHCET 2001]
(a) Classification of chemicals found in plants
(b) Use of phytochemical data in systematic botany
(c) Application of chemicals on herbarium sheets
(d) Use of statistical methods in chemical yielding plants

402. Descending arrangements of categories is called [NCERT Pg. No.8] [MHCET 2001]
(a) Classification (b) Taxonomy (c) Hierarchy (d) Key

403. Holotype is [Pb. PMT 2001]
(a) Specimen used by author as nomenclature type
(b) Specimen referred alongwith original description
(c) Duplicate of nomenclature type
(d) Specimen selected from original when nomenclature type is missing

404. Keystone species are [Pb. PMT 2001]
(a) Species belonging to same period
(b) Species that determine structure of biotic community
(c) Species reproducing sexually
(d) Species recorded only in the fossil state

405. Nicotiana is [NCERT; Haryana PMT 2001]
(a) Variety (b) Subspecies (c) Species (d) Genus

406. Barophilic prokaryotes [CBSE PMT 2005]
(a) Grow slowly in highly alkaline frozon lakes at high altitudes
(b) Occur in water containing high concentrations of barium hydroxide
(c) Grow and multiply in very deep marine sediments
(d) Readily grow and divide in sea water enriched in any soluble salt of barium

407. In hierarchical classification class is interpolated between [NCERT Pg. No.10] [Chd. CET 2002]
(a) Family and genus (b) Phylum and order
(c) Order and family (d) Kingdom and phylum

408. Binomial nomenclature is [MP PMT 1997; Chd. CET 2002] [NCERT Pg. No.6]
(a) Two words in name of a species
(b) Two names local and specific
(c) Two names of a species
(d) Two phases, asexual and sexual, in the life cycle of a species

409. A true species consists of a population [CBSE PMT 2002]
(a) Sharing the same niche (b) Interbreeding
(c) Feeding over the same food (d) Reproductively isolated

410. Species are considered as
[DPMT 1997; AIIMS 2000; BHU 2001; Kerala PMT 2002; CBSE PMT 2003; MP PMT 2009]
[NCERT Pg. No.9]
(a) Real units of classification devised by taxonomists
(b) Real basic units of classification
(c) The lowest units of classification
(d) Artificial concept of human mind which cannot be defined in absolute terms

411. Biosystematics aims at [CBSE PMT 2003]
(a) Identification and arrangement of organisms on the basis of their cytological characteristics
(b) The classification of organisms based on broad morphological characters
(c) Delimiting various taxa of organisms and establishing their relationships
(d) The dassification of organisms based on their evolutionary history and establishing their phylogeny on the totality of various parameters from all fields of study

412. Select correctly written scientific name of Mango which was first described by Carolus Linnaeus
[NEET 2019] [NCERT Pg. No.9]
(a) Mangifera indica Car. Linn. (b) Mangifera indica Linn.
(c) Mangifera indica (d) Mangifera Indica

413. As we go from species to kingdom in a taxonomic hierarchy, the number of common characteristics [NCERT Pg. No.8]
(a) Will decrease (b) Will increase (c) Remain same (d) May increase or decrease

414. Which of the following 'suffixes' used for units of dassification in plants indicates a taxonomic category of 'family' [NCERT Pg. No.8]
(a) -Ales (b) -Onae (c) -Aceae (d) -Ae

415. The term 'systematics' refers to
(a) Identification and classification of plants and animals
(b) Nomenclature and identification of plants and animals
(c) Diversity of kinds of organisms and their relationship
(d) Different kinds of organisms and their classification

416. Genus represents [NCERT Pg. No.9]
(a) An individual plant or animal
(b) A collection of plants or animals
(c) Group of closely related species of plants or animals
(d) None of these

417. The taxonomic unit 'Phylum' in the classification of animals is equivalent to which hierarchial level in classification of plants [NCERT Pg. No.10]
(a) Class (b) Order (c) Division (d) Family

418. Botanical gardens and zoological parks have [NCERT Pg. No. 12, 13]
(a) Collection of endemic living species only
(b) Collection of exotic living species only
(c) Collection of endemic and exotic living species
(d) Collection of only local plants and animals

419. Taxonomic key is one of the taxonomic tools in the identification and classification of plants and animals. It is used in the preparation of [NCERT Pg. No.13, 14]
(a) Monographs (b) Flora (c) Both (a) and (b) (d) None of these

420. All living organisms are linked to one another because [NCERT Pg. No.3, 4]
(a) They have common genetic material of the same type
(b) They share common genetic material but to varying degrees
(c) All have common cellular organization
(d) All of above

421. Which of the following is a defining characteristic of living organisms [NCERT Pg. No.5]
(a) Growth (b) Ability to make sound
(c) Reproduction (d) Response to external stimuli

422. Match the following and choose the correct option [NCERT Pg. No.9, 10]

A. Family	i. tuberosum
B. Kingdom	ii. Polymoniales
C. Order	iii. Solanum
D. Species	iv. Plantae
E. Genus	v. Solanacea

(a) i-D, ii-C, iii-E, iv-B, v-A (b) i-E, ii-D, iii-B, iv-A, v-C
(c) i-D, ii-E, iii-B, iv-A, v-C (d) i-E, ii-C, iii-B, iv-A, v-D

Critical Thinking

C_T Objective Questions C_T

423. Karyotaxonomy is the modern branch of classification which is based on [NCERT; JIPMER 1998; MP PMT 1999]
(a) Number of chromosomes (b) Bands found on chromosomes
(c) Organic evolution (d) Trinomial nomenclature

424. Which of the following combinations is correct for wheat [NCERT Pg. No.11][DUMET 2010]
(a) Genus : Triticum, Family : Anacardiaceae,

Order : Poales, Class: Monocotyledonae
(b) Genus : Triticum, Family : Poaceae,

Order : Poales, Class : Dicotyledonae
(c) Genus : Triticum, Family : Poaceae,

Order : Sapinadales, Class: Monocotyledonae
(d) Genus : Triticum, Family : Poaceae,

Order : Poales, Class: Monocotyledonae

425. Hutchinson's system of classification was revised in [DPMT 2003]
(a) 1995 (b) 1959 (c) 1954 (d) 1946

426. The total number of species included in the animal kingdom are about [CBSE PMT 1992; DUMET 2010]
(a) 1 million (b) 2 million (c) 10 million (d) 1 billion

427. Which one of the taxonomic aids can give comprehensive account of complete compiled information of any one genus or family at a particular time
[Kerala PMT 2009] [NCERT Pg. No.11]
(a) Taxonomic key (b) Flora (c) Herbarium (d) Monograph (e) Dictionary

428. A species is defined as "the group of actually or potentially inter-breeding natural population producing fertile offspring and reproductively isolated from such other groups". The above statement is given by [MP PMT 1997]
(a) Carolus Linnaeus (b) Mayr (c) J.B. Lamarck (d) Charles Darwin

429. Which of the following is required as equivalent to subspecies of classical Taxonomy [VITEEE 2006]
(a) Ecospecies (b) Ecotype (c) Cenospecies (d) Comparium

430. Which one of the following statement correctly define the term homonym [WB JEE 2008]
(a) Identical name of two different taxon
(b) Two or more names belonging to the same taxon
(c) When species name repeats the generic name
(d) Other name of a taxon given in a language other than the language of zoological / botanical nomenclature

431. Match the items given in Column I with those in Column II and select the correct option given below

Column I
(a) Herbarium
(b) (Key
(c) (Museum
(d) Catalogue

Column II
i. It is a place having a collection of preserved plants and animals
ii. A list that enumerates methodically all the species found in an area with brief description aiding identification
iii. Is a place where dried and pressed plant specimens mounted on sheets are kept
iv. A booklet containing a list of characters and their alternates which are helpful in identification of various taxa [NCERT Pg. No.11, 12] [NEET 2018]

	a	b	c	d
(a)	i	iv	iii	ii
(b)	iii	ii	i	iv
(c)	ii	iv	iii	i
(d)	iii	iv	i	ii

Latest Trend L_T Objective Questions

432. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask [NEET 2020]

(a) CH_3, H_2, NH_3 and water vapor at 600°C (b) CH_4, H_2, NH_3 and water vapor at 800°C
(c) CH_3, H_2, NH_4 and water vapor at 800°C (d) CH_4, H_2, NH_3 and water vapor at 600°C

433. According to Robert May, the global species diversity is about [NEET 2020]
(a) 7 million (b) 1.5 million (c) 20 million (d) 50 million

Assertion and Reason

`Read the assertion and reason carefully to mark the correct option out of the options given below:
(a) If both the assertion and the reason are true and the reason is a correct explanation of the assertion
(b) If both the assertion and reason are true but the reason is not a correct explanation of the assertion
(c) If the assertion is true but the reason is false
(d) If both the assertion and reason are false
(e) If the assertion is false but reason is true

Assertion : Phylogeny is the developmental history of a species.
Reason : Species is the basic unit of taxonomy.

434. Assertion : Whittaker's classification for algae is not acceptable.
Reason : Whittaker grouped algae in different kingdoms.
[NCERT Pg. No.17]
435. Assertion : Chemotaxonomy is classifying organisms at molecular level.
Reason : Cytotaxonomy is classifying organisms at cellular level.
436. Assertion : Whittaker did not include unicellular green algae in protista.
Reason : Distinction between unicellular and multicellular organisms is not possible in case of algae.
437. Assertion : Systematics is the branch of biology that deals with classification of living organisms.
Reason : The aim of classification is to group the organisms.[NCERT Pg. No.8] [AIIMS 2002]
438. Assertion : Acraniata is a group of organisms which do not have distinct cranium.
Reason : It includes small marine forms without head.
[AIIMS 1997]
439. Assertion : To give scientific name to plant, there is ICBN.
440. Reason : It uses articles, photographs and recommendations to name a plant.
[NCERT Pg. No.6] [Haryana PMT 2000]
441. Assertion : Taxon and category are same things.
Reason : Category shows hierarchical classification.
[NCERT Pg. No.8, 9]
442. Assertion : The hierarchy includes seven obligate categories.
Reason : Intermediate categories are used to make taxonomic positions more informative.
[NCERT P.No.10]
443. Assertion : The species is reproductively isolated natural population.
Reason : Prokaryotes cannot be kept under different species on the basis of reproductive isolation.

11.	Assertion		Bacteria, Protista do not have circulatory system.
	Reason	:	These organisms live in moist and watery environment.
12.	Assertion	:	Living organisms possess specific individuality with the definite shape and size.
	Reason	:	Both living and non living entities resemble each other at the lower level of organization.

Answers

Nature and Scope of Biology

1	d	2	c	3	c	4	c	5	d
6	a	7	d	8	b	9	d	10	b
11	b	12	C	13	d	14	a	15	C
16	d	17	a	18	b	19	c	20	b
21	a	22	d	23	b	24	a	25	a
26	d	27	a	28	d	29	b	30	d
31	c	32	a						

Understanding Life

1	a	2	d	3	a	4	d	5	c
6	a	7	b	8	d	9	a	10	b
11	b	12	a	13	a	14	c	15	d
16	a	17	c	18	a	19	b	20	d
21	d	22	d	23	b	24	d	25	C
26	d	27	b	28	a	29	c	30	a
31	b	32	a						

Systematics

1	a	2	a	3	d	4	a	5	b
	h	3	d	8	a	9	a	10	b
			d	13	d	14	d	15	d
			d	18	c	19	a	20	c
			d	23	d	24	a	25	b
	[redacted]		c	28	b	29	b	30	b
	c	32	8	33	c	34	a	35	d
34	b	37	a	38	d	39	a	40	c
4	d	42	c	43	d	4	b	45	a
4	d	47	d	48	8	49	C	50	b
51	1	52	*a*	53	d	64	8	55	a
	a	5	b	58	8	59	b	60	c

3	*b*	62	d	63	8	64	C	65	c

66	a	67	a	68	c	69	b	70	c
71	b	72	d	73	c	74	b	75	a
76	a	77	c	78	b	79	c	80	a
81	d	82	a	83	b	84	d	85	b
86	c	87	a	88	c	89	d	90	d
91	b	92	c	93	d	94	d	95	c
96	e	97	a	98	c	99	c	100	a
101	a	102	d	103	b	104	c	105	d
106	a	107	a	108	a	109	a	110	d
111	c	112	a	113	b	114	c	115	a
116	b	117	d	118	c	119	b	120	a
121	d	122	b	123	d	124	b		

NCERT Exemplar Questions

1	*a*	2	*c*	3	*c*	4	*c*	5	*c*
6	*c*	7	*c*	8	*d*	9	*d*	10	*a*

Critical Thinking Questions and Latest Trend

1	*b*	2	*d*	3	*a*	4	*a*	5
6	*b*	7	*c*	8	*a*	9	*d*	10
11	*a*							

Assertion and Reason

1	*b*	2	*a*	3	*b*	4	*a*	5	*b*
6	*b*	7	*a*	8	*e*	9	*b*	10	*b*
11	*b*	12	*b*						

Nature and Scope of Biology

444. (d) Edaphology is study of soil.

445. (b) Pharmacology is the branch of medicine and biology concerned with the study of drug action, where a drug can be broadly defined as any man made, natural or endogenous (with in the body) molecule which exerts a biochemical and/or physiological effect on the cell, tissue, organ or organism.

446. (c) Phenology is the scientific study of seasonal changes, i.e., periodic phenomena of organisms in relation to their climate.

Understanding Life

447. (a) In endergonic reactions, the product have more energy than the reactants, So the reaction requires an input of energy.

448. (b) Iron is component of haemoglobin of RBC of blood and helps in transportation of oxygen (about 97-99%) as well as carbon dioxide (about 23%).

449. All living organism are linked to one another because:
(1) They have common genetic material of same type.
(2) They share common genetic material but to varying degrees.
(3) All have common cellular organization.
(4) All of the above.

450. Which of the following is a defining characteristic of living organism?
(1) Growth
(2) Ability to make sound
(3) Reproduction
(4) Response to external stimuli

451. Metabolism is a feature of:
(1) Living beings.
(2) Nonliving beings.
(3) Both (1) and (2).
(4) Virus.

452. Living and nonliving objects of nature:
(1) Have similar materials.
(2) Obey same physiochemical laws.
(3) Both (1) and (2).
(4) Have same materials, but obey different physiochemical and biochemical laws.

453. Which one of the following is a defining property of living being?
(1) Growth
(2) Reproduction
(3) Metabolism
(4) Unconsciousness

454. Apoplasmic growth is through the formation of:
(1) Cells walls and cell junctions.
(2) Matrix and fibers of connective tissue.
(3) Intake of water.
(4) Both (1) and (2).

455. Homeostasis can be defined as:
(1) Tendency of the animals to change with the surroundings.
(2) Tendency of the living organism to change the external environment as per their requirements.
(3) Tendency of the living organism to resist changes in the external environment.
(4) Tendency of the biological systems to resist changes.

456. The living organisms can be unexceptionally distinguished from the nonliving things on the basis of their ability for:
(1) Reproduction.
(2) Growth and movement.
(3) Responsiveness to touch.
(4) Interaction with environment and progressive evolution.

457. Mule is the hybrid produced under captive conditions by the cross of:
(1) Female donkey and male horse.
(2) Male donkey and female horse.
(3) Male tiger and female lion.
(4) Female tiger and male lion.

458. Extrinsic growth can be seen in:
A. Mountains. B. Animals. C. Viruses. D. Boulders.
E. Sand mounds. F. Rolling ice.
(1) A, D, E, F (2) B, C, D, E (3) C, D, E, F (4) A, B, D, E

459. Process of regeneration is seen in:
(1) Fungi. ` (2) Yeasts. (3) Algae. (4) Planaria.

460. Self-consciousness is a feature of:
(1) Only human beings. (2) All animals.
(3) All organisms. (4) All plants.

461. Which of the following are considered as neither living nor nonliving?
A. Coma patients B. Prions
C. Viroids D. Virions
E. Mycoplasmas F. Ray fungi
(1) A, B, D, and F (2) B, C, E, and F (3) A, B, C, and D (4) A, B, C, and E

462. Which of the following is incorrect?
(1) Cellular organization and consciousness are defining properties of living beings.
(2) Systematics involves taxonomy along with phylogeny.
(3) Trinomial nomenclature system was proposed by Caesalpino and Bauhin.
(4) Scientific names of cultivated plants have been standardized through ICNCP.

463. Which of the following is not the defining properties of living beings?
(1) Intrinsic growth (2) Sexual reproduction
(3) Cellular organization (4) Irritability

464. The most obvious and technically complicated feature of all living organisms is:
(1) Reproduction. (2) Growth.
(3) Ability to sense their environment. (4) Photosynthesis.

465. Which of the following character is not common to both living and nonliving?
(1) Growth (2) Metabolism (3) Both (1) and (2) (4) None of these

466. The term systematics refers to:
(1) Identification and classification of organisms.
(2) Nomenclature and identification of organisms.
(3) Diversity of kinds of organisms and their relationship.
(4) Different kinds of organisms and their classification.

467. New systematics introduced by Sir Julian Huxley is also called as:
(1) Phenetics. (2) Cladistics.
(3) Biosytematics. (4) Numerical taxonomy.

468. Bionomial nomenclature was published in:
(1) Systema Naturae. (2) Genera Plantarium.
(3) Genera Animalium. (4) Historia Plantarium.

469. ICZN was adopted in:
(1) 1960 . (2) 1970 . (3) 1964 . (4) 1974.

470. "Systema Naturae" was written by:
(1) Linnaeus. (2) Aristotle. (3) Hippocrates. (4) Darwin.

471. Autonym is valid in:
(1) Only in botanical nomenclature.
(2) Only in zoological nomenclature.
(3) Both in botanical and zoological nomenclature.
(4) Only in bacteriological nomenclature.

472. "Taxonomy without phylogeny is like bones without flesh" was remarked by:
(1) John Hutchinson. (2) Takhtajan.
(3) Oswald Tippo. (4) Bentham and Hooker.

473. The classification of organisms on the basis of number and size of chromosomes is called as:
(1) Biochemical taxonomy. (2) Numerical taxonomy.
(3) Chemotaxonomy. (4) Karyotaxonomy.

474. Basic unit of taxonomy is:
(1) Class. (2) Order. (3) Genus. (4) Species.

475. ICNCP is:
(1) International code for nomenclature of cultivated plants.
(2) Indian code for nomenclature of cultivated plants.
(3) International code for natural conservation project.
(4) International code for naming conserved plants.

476. Binomial nomenclature for plants became effective from:
(1) 06-07-1736. (2) 01-05-1753. (3) 01-08-1758. (4). 05-08-1771.

477. Aristotle divided animals into:
(1) Protozoa and metazoan. (2) Vertebrata and invertebrate.
(3) Chordata and nonchordata. (4) Enaima and Anaima.

478. Rattus rattus represents;
(1) Autonyms. (2) Homonyms. (3) Tautonyms. (4) Synonyms.

479. The book Systema Naturae was written by:
(1) Lamarck. (2) Cuvier. (3) Aristotle. (4) Linnaeus.

480. Which of the following name is correct?
(1) Apis indica (2) Trypasoma gambiense
(3) Ficus Bengal (4) Mangifera Indica

481. The third name of the trinomial nomenclature is of:
(1) Subgenus (2) Species (3) Subspecies (4) Author

482. Corvus splendens splendens is the scientific name of Indian Crow. It represents:
(1) Binomial name. (2) Tautonym. (3) Autonym. (4) Synonym.

483. Systematics is the study of:
(1) Diversity amongst groups of organisms.
(2) Grouping of organisms.
(3) Identification and grouping of organisms.
(4) Identification, classification, and taxonomy.

484. In which one of the following systems each character is given equal importance and at the same time hundreds of characters can be considered?
(1) Morphotaxonomy (2) Numerical taxonomy
(3) Cytotaxonomy (4) Chemotaxonomy

485. Concept of three domains was given by:
(1) Whittaker (2) Linnaeus. (3) Copeland. (4) Carl Woese.

486. A natural classification of organisms was given by:
(1) Theophrastus. (2) Bentham and Hooker.
(3) Linnaeus. (4) All of these.

487. uses the chemical constituents of the plants to classify them.
(1) Cytotaxonomy (2) Karyotaxonomy
(3) Chemotaxonomy (4) Numerical taxonomy

488. Artificial system of classification given by Linnaeus considers:
(1) Vegetative characters.

(2) Chemical characters.
(3) Floral characters.
(4) The characters more affected by environment.

489. Each character is given equal weightage and at the same time hundreds of characters can be considered in:
(1) Numerical taxonomy. (2) Cytotaxonomy.
(3) Chemotaxonomy. (4) Morphotaxonomy.

490. Four-Kingdom system of classification was given by:
(1) Haeckel. (2) Copeland. (3) Whittaker. (4) John Ray.

491. Which of the following was not a kingdom in fourKingdom system of classification?
(1) Monera (2) Protista (3) Fungi (4) Plantae

492. Three-Kingdom system of classification was given by:
(1) Haeckel. (2) Copeland. (3) Whittaker (4) John Ray.

493. A phylogenetic system of classification was proposed by:
(1) Hutchinson. (2) Engler and Prantl. (3) Lamarck. (4) Both (1) and (2).

494. Which one of the following systems of classification assumes that organisms belonging to the same taxa have a common ancestor?
(1) Earliest system (2) Artificial system
(3) Natural system (4) Phylogenetic system

495. Phenetic classification of organisms is based on:
(1) Observable charactetistics of existing organisms.
(2) The ancestral history of existing organisms.
(3) DNA characteristics.
(4) Sexual characteristics.

496. DNA sequence is the basis of grouping of organisms in:
(1) Artificial system. (2) Cytotaxonomy.
(3) Phenetics. (4) Chemotaxonomy.

497. Phylogenetic system of classification is based on:
(1) Biochemical properties. (2) Morphological similarities.
(3) Evolutionary relationships. (4) Chemical characteristics of organisms.

498. An example for artificial system of classification is:
(1) Bentham and Hooker system. (2) Linnaeus system.
(3) Engler and Prantl system. (4) Hutchinson system.

499. The term "New systematics" was introduced by:
(1) Linnaeus. (2) AP Candolle. (3) Julian Huxley. (4) Bentham and Hooker.

500. In vitro, metabolic reactions means:
(1) Reactions that occur in laboratory conditions.
(2) Reactions that occur in cell-free system.
(3) Both (1) and (2).
(4) None of the above.

501. Ficus benghalensis is the scientific name of:
(1) Peepal. (2) Fig. (3) Banyan. (4) Neem.

502. Biosystematics aims at:
(1) Identification and arrangement of organisms on the basis of their cytological characteristics.
(2) The classification of organisms based on broad morphological characters.
(3) Delimiting various taxa of organisms and establishing their relationships.
(4) The classification of organisms based on their evolutionary history and establishing their phylogeny on the totality of various parameters from all fields of study.

503. Founder of Binomial nomenclature was:
(1) Linnacus. (2) Mendel. (3) Darwin. (4) Lamarck.
504. First act in taxonomy is:
(1) Description. (2) Identification. (3) Naming. (4) Classification.
505. Which statement is true?
(1) Tautonyms are not allowed in plants.
(2) Tautonyms are not allowed in animals.
(3) Tautonyms are normally allowed in animals and sometimes allowed in plants.
(4) Tautonyms are allowed only in bacteria.
506. Which of the following is a correct name?
(1) Solanum Tuberosum (2) Solanum tuberosum
(3) Solanum tuberosum linn (4) All of the above.
507. Find odd one with respect to taxon.
(1) Gourds (2) Palms (3) Conifers (4) Grasses
508. Match the following and choose the correct option.Column - I
A. Family B. Kingdom C. Order D. Species E. Genus

	i	ii	iii	iv	v		i	ii	iii	iv
(1)	D	C	E	B	A	(2)	E	D	B	A
(3)	D	E	B	A	C	(4)	E	C	B	A
(3)	D									

509. Species as a unit of classification was first employed by:
(1) Huxley. (2) De Candolle. (3) John Ray. (4) Linnaeus.
510. Order primata and carnivora are placed in the same class:
(1) Hominidae. (2) Mammalia. (3) Insecta. (4) Chordata.
511. A genus with a single species is:
(1) Monotypic. (2) Typical. (3) Atypical. (4) Polytypic.
512. Common and generic names are similar in case of:
(1) Felis. (2) Gorilla. (3) Mangifera. (4) Saccharum.
513. Taxon Tiger represents:
(1) Species., (2) Genus. (3) Family. (4) Glass.
514. Roundworm is a taxon that denotes:
(1) Genus. (2) Family, (3) Phylum. (4) Class.
515. The suffix phyta indicates:
(1) Family. (2) Order. (3) Class. (4) Division.
516. In a hierarchial system of plant classification, which one of the following taxonomic ranks generally ends in "aceae"?
(1) Family (2) Genus (3) Order (4) Class
517. Panthera is a:
(1) Variety. (2) Subspecies. (3) Species. (4) Genus.
518. The correct sequence of the various taxons while naming an organism is:
(1) Order, class, family. (2) Class, order, family.
(3) Family, class, order. (4) Order, family, class.
519. The intermediate category between subfamily and genus is:
(1) Race. (2) Clone. (3) Tribe. (4) Subspecies.
520. Mango belongs to class dicotyledonae of angiosperms and belongs to order:
(1) Poales. (2) Sapindales. (3) Ranales. (4) Parietales.

521. A group having interbreeding capacity and ability to produce fertile offsprings is:
(1) Community. (2) Species. (3) Class. (4) Genus.
522. Polymoniales is an (category):
(1) Order. (2) Class. (3) Division. (4) Subclass.
523. Choose the incorrect statement.
(1) Higher the taxonomic category, higher the number of organisms.
(2) Higher the taxonomic category, lower the number of organisms.
(3) Higher the category, lower will be the number of common characters.
(4) There are seven obligate categories.
524. Basic unit of classification is:
(1) Genus. (2) Species. (3) Order. (4) Class.
525. Taxonomic Aids77. Which of the following taxonomical aid is related with quick referral systems in taxonomic studies?
(1) National Zoological Park, Delhi (2) NMNH, Delhi
(3) Central National Herbarium, Calcutta (4) Both (1) and (2)
526. Select the odd one out with respect to botanical gardens.
(1) Collection of living plants for reference
(2) Indian Botanical Garden is at Howrah
(3) It is a method of ex situ study
(4) Collection of preserved plants and animals specimens
527. Read the following statements carefully and select the correct option.
I. In zoological parks, conditions similar to natural habitats are provided to animals.
II. Keys are generally analytical in nature.
III. In herbarium sheet, local names are not mentioned.
IV. Taxonomical aids are useful in knowing the biorewources.
(I) Only (I) and (II) (2) (I), (III), and (IV)
(3) (I), (II), and (IV) (4) All of the above
528. Which of the following includes the alphabetical arrangement of species of a particular place describing features?
(1) Manuals (2) Catalogues (3) Monograph (4) Flora
529. Match column-I with column-II

Column I	Column II
a. Museum	(i) Information on one taxon
b. Herbaria	(ii) Couplet
c. Botanical garden	(iii) Arranged on universally accepted classification system
d. Taxonomic key	(iv) Educationalinstitutes
	(v) Records oflocal flora for monographic work

(1) a (iv), b (iii), c (v), d (ii) (2) a (ii), b (v), c (i), d (ii)
(3) a (iv), b (i), c (iii), d (v) (4) a (v), b (iii), c (iv), d (ii)
Select the incorrect statement.
(1) Museums often have collections of skeletons.
(2) Separate taxonomic keys are required for each taxonomic category.
(3) Taxonomic keys are based on the contrasting characters.
(4) Monograph is useful in providing information for identifications of names of species found in an area.
530. Study the following statements regarding the preparation of herbarium sheets.
I. Plant should be collected in the flowering stage.
II. Every detail regarding the plant such as locality, ecological conditions, vegetative, and floral

characters, etc., should be noted.
III. Plants are evenly pressed by unfolding all the plant parts between blotting papers (or newspapers) with the help of plant press.
IV. Blotting papers need not be changed, until the plant gets dried.
V. After drying, the plant specimen is carefully mounted/ pasted on the herbarium sheets.
VI. The herbarium sheet is labeled on the lower right-hand corner representing the number of plant specimen, date of collection, etc.

531. Which of the above statements is/are not correct?
(1) I only (2) IV only (3) I and IV (4) III and IV

532. Read the following statements regarding biological museums.
I. Biological museums are generally set up in educational institutes such as schools and colleges.
II. Museums have collections of preserved plant and animal specimens for study and reference.
III. Specimens are preserved in the containers of jars in preservative solutions.
IV. Insects are preserved in insect boxes after collecting, killing, and pinning.
V. Larger animals like birds and mammals are usually stuffed and preserved.
VI. Skeletons of mammals are not allowed to be kept in museums.

533. Which of the above statements is/are not correct?
(1) II and III (2) I and VI (3) V only (4) VI only

534. An example of ex situ conservation is:
(1) National park. (2) Seed bank. (3) Wildlife sanctuary. (4) Sacred groove.

535. Largest herbarium of India is at:
(1) Lloyd Botanical Garden, Darjeeling. (2) Indian Botanical Garden, Sibpur.
(3) National Botanical Garden, Lucknow. (4) Forest Research Institute, Dehradun.

536. Type specimen described as the nomenclature type is:
(1) Isotype. (2) Paratype. (3) Syntype. (4) Holotype.

537. The biological concept of species was given by:
(1) E Mayr. (2) Linnaeus. (3) Lamarck. (4) RH Whittaker.

538. Two apparently similar populations are intersterile. They are members of:
(1) One species. (2) Two biospecies. (3) Two subspecies. (4) Two extinct species.

539. Why reproduction is not an inclusive defining characteristic of all living organisms?
(1) Nonlivings beings also reproduce. (2) Many living organisms are sterile.
(3) Reproduction is synonym to growth. (4) Both (1) and (2).

540. Choose the incorrect statement regarding universal rules of nomenclature.
(1) The first word of a biological name represents the genus.
(2) The first word denoting the genus starts with a capital letter.
(3) Both the words in a biological name, when handwritten, are separately underlined.
(4) Biological names are generally in English and are printed in italics.

541. A taxon is a:
(1) Group of related species. (2) Group of related families.
(3) Type of living organisms. (4) Taxonomic group of any ranking.

542. Solanum and Panthera are:
(1) Genus and species. (2) Genus and genus.
(3) Species and species. (4) Only species.

543. According to five-kingdom system of classification, how many kingdoms contain multicellular eukaryotes?
(1) Three kingdoms (2) Four kingdoms
(3) Two kingdoms (4) All the five kingdoms

544. "Taxa" differs from "taxon" due to:
(1) This being a higher taxonomic category than taxon.
(2) This being a lower taxonomic category than taxon.
(3) This being the plural of taxon.
(4) This being the singular of taxon.

545. Which one of the following represents a tautonym?
(1) Rattus rattus (2) Prumus dulcis
(3) Corvus splendens splendens (4) Albugo candida

546. Corvus splendens splendens is the scientific name of Indian cow. It represents:
(1) Binomial nomenclature. (2) Autonyms.
(3) Tautonyms. (4) Synonyms.

547. Type of specimen described along with holotype is:
(1) Isotype. (2) Paratype. (3) Topotype: (4) Syntype.

548. Reproduction is synonymous with growth of which group of organisms?
(1) Multicellular filaments organisms (2) Colonial organisms
(3) Unicellular organisms (4) All of the above

549. The word species was coined by:
(1) Aristotle. (2) Linnaeus. (3) John Ray. (4) Engler.

550. Go through the given statements regarding an order and choose the codes with all correct statements.
I. It is an assemblage of families.
II. It contains less similar characters than genera.
III.It is higher in systematic hierarchy than class.
IV. It is higher in systematic hierarchy than phylum.
Codes:
(1) I, II, and IV (2) I and II (3) II and IV ` (4) II and III

551. Match the following columns.

Column I	Column II
A. Family	Tuberosum
B. Kingdom	Polynomiales
C. Order	Solanum
D. Species	Plantae
E. Genus	Solanaceae

552. Which one of the following is not a correct statement?
(1) Herbarium houses dried, pressed, and preserved plant specimens.
(2) Botanical gardens have collection of living plants for reference.
(3) A museum has collection of photographs of plants and animals.
(4) Key is a taxonomic aid for identification of specimens.

553. Read the statements given below and identify the incorrect statement.
(1) Scientific names inhibit multiple naming for the same kind of an organism.
(2) Scientific names are used all over the world.
(3) Scientific names are often descriptive and tell us some important characters of an organism.
(4) Scientific names indicate relationship between species.

554. The growth by cell division occurs continuously throughout their life span in:
(1) Majority of lower animals. (2) Higher animals.
(3) Plants. (4) Only heterotrophic plants.

555. The filamentous algae and the protonema of moss easily multiply by:
(1) Spores formation. (2) Fragmentation.
(3) Budding. (4) Binary fission.

556. Growth and reproduction are mutually exclusive events for:
(1) Monerans.
(2) Protists.
(3) Plants and few bacteria.
(4) Majority of higher plants and animals.

557. Which one of the following cannot be taken as defining property of all life forms?
(1) Cellular organization
(2) Self-consciousness
(3) Ability to interact and respond to environmental stimuli
(4) Metabolism

558. Isolated metabolic reactions in vitro are:
(1) Living things.
(2) Nonliving things.
(3) Living reactions.
(4) More than one option is correct.

559. Consider the given statements.
A. Nonliving objects grow by increase in mass.
B. Properties of tissue arise as a result of interactions among the constituent cells.
(1) Both A and B are correct.
(2) Only *A* is correct.
(3) Only B is correct.
(4) Both A and B are incorrect.

560. For animals, scientific names are based on agreed principles and criteria, which are provided in:
(1) ICBN.
(2) ICNB.
(3) ICZN.
(4) ICTV.

561. Which one of the following is a correct representation of binomial epithet with respect to mango?
(1) Mangifera Indica
(2) Mangifera indica Linn.
(3) Mangifera Indica Linn.
(4) Mangifera indica L.

562. In the given statement "potato is a dicot and dicots are plants".
(1) Taxon potato represents family
(2) Taxon dicot represents class
(3) Taxon plant represents species
(4) Taxon potato represents genus

563. Taxonomic category with comparatively less number of common characters is:
(1) Family.
(2) Order.
(3) Genus.
(4) Variety.

564. Taxonomic category with a group of related genera is characterized on the basis of:
(1) Cell structure.
(2) Vegetative features.
(3) Vegetative and reproductive features.
(4) Only floral characters.

565. The animal order "Carnivora" includes families like:
(1) Felidae and Cancidae.
(2) Felidae and Hominidae.
(3) Hominidae and Cancidae.
(4) Muscidae and Hominidae.

566. The prime source of taxonomic studies is:
(1) Classification of plants and animals.
(2) Nomenclature.
(3) Identification.
(4) Collection of actual specimens of plant and animal species.

567. To facilitate more sound and scientific placement of various taxa, taxonomists have also developed in the hierarchy.
(1) Broad categories
(2) Obligate categories
(3) Subcategories
(4) More than one option is correct

568. The taxonomical aid as store house of collected plant specimens that are dried, pressed, and preserved on sheets is:
(1) Museum.
(2) Botanical gardens.
(3) Herbarium.
(4) Flora.

569. Each plant is labeled indicating its botanical name and family name only in:
(1) Botanical gardens.
(2) Museum.
(3) Herbarium.
(4) More than one option is correct.

570. Select incorrect statement for one of the taxonomical aid based on the contrasting characters generally in pair.
(1) Generally analytical in nature.
(2) Same taxonomic key required for each category such as family, genus, and species.
(3) Each statement is a lead.
(4) This results in the acceptance of only one statement out of pair.

571. Ex situ conservation of organisms is associated with:
(1) Botanical gardens and museum
(2) Herbarium and zoos.
(3) Zoos and botanical gardens.
(4) Museum and herbarium.

572. Solanum melongena and Mangifera indica belong to:
(1) Different family but same order.
(2) Different order and class.
(3) Same class and division.
(4) Different class but same division.

573. Correct sequence of taxonomical categories in descending order is:
(1) Genus → Family → Order → Class
(2) Species → Genus → Order → Division
(3) Order → Family → Genus → Species
(4) Class → Order → Tribe → Family

574. Growth is synonymous with reproduction in:
(1) Bacteria and Protists.
(2) Planaria and Spirogyra.
(3) Ferns and protonema of moss.
(4) Higher plants and animals.

575. From the given list, how many characters can be taken as defining property for all life forms?
Extrinsic growth, Reproduction, Isolated metabolic reactions in vitro, Intrinsic growth, Self-consciousness, Ability to sense only physical stimulus and mount suitable response.
(1) Three (2) Two (3) Four (4) One

576. The most obvious and technically complicated feature of all living organisms is:
(1) Growth. (3) Consciousness. (2) Reproduction. (4) Metabolism.

577. Read the statements carefully and select the appropriate answer.
A. Properties of tissues are present in the constituent cells.
B. Flat worms show true regeneration.
C. Nonliving objects also grow by increase in mass.
(1) All statements are correct.
(2) A and B are correct.
(3) A and C are incorrect.
(4) Only A is incorrect.

578. The number of species that are known and described on earth refer to (i), which ranges between (ii).
(1) (i) Classification, (ii) 1.7-1.8 billion
(2) (i) Biodiversity, (ii) 17-18 lakh
(3) (i) Taxonomy, (ii) 1.7-1.8 million
(4) (i) Biodiversity, (ii) 17-18 million

579. For plants, scientific names are based on agreed principles and criteria that are provided in:
(1) ICBN. (2) ICNB. (3) ICZN. (4) ICTV.

580. Find one with respect to taxa belonging to different levels in hierarchy.
(1) Lion, Tiger, Leopard
(2) Wheat, Monocots, Plants
(3) Felis, Panthera, Homo
(4) Dicotyledonae, Monocotyledonae

581. Select incorrect statement with respect to binomial nomenclature.
(1) Given by Carolus Linnaeus.
(2) Being practiced by biologists all over the world.
(3) First word in a biological name represents the genus, whereas the second component denotes the specific epithet.
(4) Biological names are written in Latin and underlined separately when printed.

582. Taxonomy includes:
(1) Characterization.
(2) Identification.
(3) Evolutionary relationship.
(4) More than one option is correct.

583. A group of related genera with still less number of similarities ends with suffix (according to ICBN).
(1) "ales." (2) "phyceae." (3) "phyta." (4) "aceae."

584. A group of individual organisms with fundamental similarities:
A. Is the lowest obligate category in hierarchy.
B. Cannot interbreed naturally.
(1) A and B are correct. (2) A and B are incorrect.
(3) Only B is incorrect. (4) Only A is incorrect.

585. Mark the mismatched pair.
(1) Indian Botanical Garden – Howrah (2) Systema Naturae - Linnaeus
(3) Autotrophic moneran – Anabaena (4) Monotypic genera – Solanum

586. Plant families like Convolvulaceae and Solanaceae are included in the:
(1) Different orders and same class (2) Same order and different classes
(3) Same order as well as same class (4) Different orders and classes

587. In the given box, how many taxonomical aids are concerned with the conservation of live specimens for future studies? Zoological parks, Museum, Botanical gardens, Herbarium
(1) One (2) Two (3) Three (4) Four

588. Select the correct match:

Column-I	Column-II
a. Ex situ conservation of plants	(i) Herbaria
b. Serve as quick referral systems in taxonomical	(ii) NBRI, Lucknow studies
c. Collection of skeletons of	(iii) Zoos animals too
d. Enable us to learn about foot habits and behavior of animals	(iv) Museum

(1) a (ii), b (i), c (iv), d (iii) (2) a (ii), b (i), c (iii), d (iv)
(3) a (i), b (ii), c (iv), d (iii) (4) a (iv), b (i), c (iii), d (ii)

589. Which one of the following represents the choice made between two opposite options which results in acceptance of only one and rejection of the other?
(1) Flora (2) Manuals (3) Key (4) Monographs

590. The actual account of habitat and distribution of plants of given area is given in:
(1) Flora. (2) Catalogues. (3) Manuals. (4) Monographs.

591. Select incorrect statement with respect to Aristotle system of classification.
(1) First to attempt a more scientific basis for classification.
(2) It was given on the basis of usage of various organisms and mode of nutrition.
(3) Used simple morphological characters to classify plants into three groups.
(4) Also divided animals on the basis of presence and absence of red blood cells.

592. Carolus Linnaeus is credited for all, except:
(1) Coining the term systematics.
(2) Giving binomial nomenclature.
(3) Developing two-kingdom system.
(4) Using main criterion "mode of nutrition" for classifying the organisms.

593. Nostoc, Chlamydomonas, Yeast, and Spirogyra are placed together in the same kingdom in (A) system mainly on the basis of
(1) Two kingdom, a)Cell wall
(2) Four kingdom, b)Cell type
(3) Two kingdom, c)Body organization
(4) Five kingdom, d)Body organization

594. Which one of the following character was not used by Whittaker in his classification system?
(1) Cell structure (2) Thallus organization
(3) r-RNA gene sequencing (4) Phylogenetic relationship

595. Whittaker classification system did not differentiate which of the following group of organisms as kingdom?
(1) Fungi and plants (2) Animals and bacteria
(3) Eubacteria and archaebacterial (4) Both (2) and (3)

596. Organisms with cellular body organization only and loose tissue level body organization are placed into total (with respect to Whittaker's system)
(1) One and one kingdom, respectively. (2) One and two kingdom, respectively.
(3) Two and one kingdom, respectively. (4) Two and two kingdom, respectively.

597. Select odd one with respect to broad categories in taxonomic hierarchy.
(1) Kingdom and Division (2) Phylum and Class
(3) Subspecies and Tribe (4) Order and Family

598. Which of the following functions can be performed by taxonomic aids?
(1) Identification, Store house (2) Classification, Naming
(3) Conservation, Preservation (4) More than one option is correct

599. Choose the correct option for taxonomic keys.
I. Each statement of key is A .
II. Set of characters used in a pair is B .
III. Pairs of C characteristics are used.
(1) A - Couplet, B - Lead, C - Non-alternating
(2) A - Lead, B - Couplet, C - Non-contrasting
(3) A - Lead, B - Couplet, C - Alternate
(4) A - Couplet, B - Lead, C - Contrasting

600. Which of the following serve as means for ex-situ conservation of rare and endemic plant and animal species respectively?
(1) Herbarium, Museum (2) Botanical Garden, Zoological park
(3) Botanical Garden, Museum (4) Herbarium, Zoological park

601. Higher the taxonomic category, is the number of organisms and is the number of common features.
(1) Higher, more (2) Fewer, less (3) Higher, fewer (4) Fewer, more

602. Choose odd one with respect to obligate categories.
(1) Genus and Family (2) Family and Order
(3) Tribe and Variety (4) Order and Class

603. The science that deals with diversity of organisms and all their comparative and evolutionary relationship is:
(1) Wider field of science than taxonomy. (2) Referred to as systematics.
(3) Based on phytogeny. (4) More than one option is correct.

604. In taxonomic hierarchy, given below organisms are representative of how many families?
Wheat, Potato, Brinjal, Mango
(1) Two (2) Three (3) Four (4) One

605. Modern taxonomic studies are similar to classical one in dealing with:
(1) Cell structure and function. (2) Internal and external features.
(3) Developmental and ecological processes. (4) Morphological features.

606. Choose incorrect option with respect to some important rules of binomial nomenclature.
(1) The generic name is followed by specific epithet and then the name of author.
(2) Both words in biological component are separately underlined when handwritten.
(3) Biological names are generally in Latin and written in italics.
(4) The name of the author is printed in Roman and underlined.

607. Consciousness is a feature and selfconsciousness is a feature of all life forms

(1) Defining, defining (2) Nondefining, nondefining

(3) Nondefining, defining (4) Defining, nondefining

608. Choose incorrect statement.

(1) No nonliving object exhibits metabolism.

(2) Metabolic reactions can occur in vito.

(3) Metabolism is a nondefining feature.

(4) Metabolic reactions can be demonstrated in cell-free system.

609. For bacteria, we are not very clear about the usage of terms like:

(1) Growth and metabolism. (2) Consciousness and cellular organization.

(3) Growth and reproduction. (4) Metabolism and reproduction.

610. Which of the following provide index to the plant species found in particular area?

(1) Manual (2) Flora (3) Monograph (4) Catalogue

611. For plants, scientific names are based on agreed principles and criteria that are provided in:

(1) ICZN. (2) ICNB. (3) ICBN. (4) ICTV.

612. Biological classification system that was introduced by Carl Woese divided the Monera into:

(1) Three domains. (2) Two domains. (3) Six domains. (4) One domain.

613. According to five-kingdom classification system below, organisms or groups belong to how many kingdoms?

Chlorella, Chlamydomonas, Mycoplasma, Archaebacteria, Amoeba, Moulds, Paramoecium

(1) Two (2) Three (3) Four (4) Five

614. Which of the following represents the correct positioning of organisms according to Linnaean classification system? (where, P: Plantae and A: Animalia)

(a) Mosses (b) Blue green algae (c) Fungi (d) Bacteria

(e) Protozoa (f) Vertebrates (g) Invertebrates

(1) P-a, b, c, d (2) P-a,b,d A-e,f,g A-c, e, f, g

(3) P-a,b,c A-d, e, f, g (4) P-a, c, e A-b, d, f, g

615. Awareness of the surroundings and response to external stimuli is:

(1) Shown by the most advance and complex eukaryotes only.

(2) Homeostasis.

(3) Shown by higher plants and animals only.

(4) Defining property of living organisms.

616. Choose incorrect option with respect to features of living organisms.

(1) Growth is seen only up to a certain age in animals.

(2) Sum total of all the chemical reactions occurring in our body is metabolism.

(3) No nonliving object is capable of reproduction.

(4) In multicellular organisms, we are not clear about the usage of terms like growth and reproduction.

617. Which of the following taxonomic categories are common for Triticum and Mangifera?

(1) Division, kingdom (2) Division, class

(3) Order, class (4) Family, order

618. Taxonomic category with a group of related genera is:

(1) Anacardiaceae, Primata. (2) Poaceae, Diptera.

(3) Hominidae, Canidae. (4) Sapindales, Anacardiaceae.

619. In classical taxonomy species are delimited on the basis of:

(1) Physiology, cytology. (2) Ecology, genetics.

(3) Morphology, external features. (4) Biochemistry, genetics.

620. Which of the following is correct with respect to rules of binominal nomenclature?
(1) Same specific name can be given to organisms belonging to different genera.
(2) Scientific name is italicized in hand written description.
(3) Specific name is followed by generic name.
(4) Second component in biological name starts with a capital letter.

621. Systematics is different from taxonomy in dealing with:
(1) Identification. (2) Classification. (3) Phylogeny. (4) Nomenclature.

622. Choose correct option for the missing words.
Solanum +A → Solanaceae
Panthera +B → C
(1) A-Lycopersicon, B-Canis, C-Canidae (2) A - Datura, B - Felis, C - Felidae
(3) A - Mangifera, B - Canis, C – Canidae (4) A - Oryza, B - Felis, C – Felidae

623. Which of the following group of organisms represent taxa at same level?
(1) Monocots, angiosperms (2) Lion, animals
(3) Wheat, mango (4) Tiger, chordates

624. Taxonomic keys are generally and are based on characters.
(1) Nonanalytical, contrasting (2) Analytical, noncontrasting
(3) Nonanalytical, noncontrasting (4) Analytical, contrasting

625. Detailed information about a particular taxon is present in:
(1) Manual. (2) Flora. (3) Monograph. (4) Catalogue.

626. Which of the following taxonomic aids provide information about the local flora as well as flora of distant areas?
(1) Museum, herbarium (2) Herbarium, botanical garden
(3) Botanical garden, museums (4) Herbarium, flora

627. Largest botanical garden of Asia is situated in .
(1) NBG, Lucknow (2) IBG, Kolkata
(3) RBG, London (4) MBG, Moscow

628. For bacteria scientific names are based on agreed principles and criteria, which are given in:
(1) ICTV. (2) ICBN. (3) ICNB. (4) ICZN.

629. Arrangement of organisms into convenient categories on the basis of similarities and differences in certain easily observable but fundamental characters is:
(1) Taxonomy. (2) Classification. (3) Nomenclature. (4) Identification.

630. Living organisms are characterized by all, except:
(1) Metabolism. (2) Ability to self-replicate.
(3) Interaction with only physical stimulus. (4) Ability to self-organize.

631. Read the statements carefully and select correct set of statements.
(a) Intrinsic growth is defining property of living beings.
(b) Yeast and Hydra reproduce asexually by spore formation.
(c) Reproduction is synonymous with growth in Amoeba.
(1) All options (2) (a) and (b) (3) (a) and (c) (4) (b) and (c)

632. The most obvious and technically complicated feature is A__in B .
(1) A - Consciousness; B - All living organisms
(2) A - Metabolism; B - All living organisms
(3) A - Self-consciousness; B - Multicellular organisms
(4) A- Reproduction; B- All organisms

633. In binomial nomenclature:
(1) Each name has two components of italics origin.
(2) Both words are generally in Latin and written in italics.
(3) The first word represents the specific epithet.
(4) Name of the author appears after the specific epithet and should be underlined if handwritten.

634. Taxonomy involves all, except:
(1) Characterization and identification of organisms.
(2) Nomenclature of organisms.
(3) Classification of organisms.
(4) Evolutionary relationships between organisms.

635. Which of the following is concerned with diversity of organisms and relationships among them?
(1) Classification (2) Taxonomy
(3) Classical taxonomy (4) Systematics

636. Choose odd one with respect to obligate categories in taxonomic hierarchy.
(1) Kingdom and Phylum (2) Class and Order
(3) Variety and Domain (4) Genus and Species

637. Biological concept of species was given by and based upon isolation.
(1) Emst Mayr, reproductive (2) Linnaeus, reproductive
(3) Emst Mayr, morphological (4) de Candolle, behavioral

638. Growth and reproduction are mutually exclusive events for:
(1) Bacteria. (2) Protistans.
(3) All unicellular and multicellular organisms. (4) Higher animals and plants.

639. More common characters are present in which one of the following taxonomic categories?
(1) Family (2) Genus (3) Order (4) Class

640. Sweet potato and potato are placed in the:
(1) Same family and different orders. (2) Different families and different orders.
(3) Different orders but same class. (4) Different families but same order.

641. Select incorrect matching with respect to taxonomic categories of wheat.
(1) Genus – Triticum (2) Family - Poaceae
(3) Order – Polymoniales (4) Division – Angiospermae

642. The number of species that are known and described range between:
(1) 1.7 and 1.8 billion (2) 1.7 and 1.8 million
(3) 17 and 18 million (4) 5 and 30 million

643. Petunia and Datura are placed in the family Solanaceae on the basis of:
(1) Vegetative feature. (2) Reproductive feature.
(3) Habit and color. (4) Both (1) and (2).

644. Find out the correct sequence of taxonomic categories in ascending order.
(1) Species → Family → Genus (2) Kingdom → Class → Division
(3) Order → Class → Division (4) Genus → Class → Order

645. Carnivore is assemblage of:
(1) Mammalia and Felidae. (2) Felidae and Cancidae.
(3) Convolvulaceae and Solanaceae. (4) More than one option is correct.

646. A family tree based upon phenetic classification is known as:
(1) Cladogram. (2) Dendrogram. (3) Cladistics. (4) Hologram.

647. Rules regarding the binomial nomenclature are framed by:
(1) The scientists of Central America. (2) Director of Royal Botanical Gardens, Kew.
(3) The International Science Congress. (4) Individuals of all the countries.

648. Which of the following category is not true for wheat?
(1) Poaceae (2) Sapindales (3) Monocotyledonae (4) Triticum
649. Which of the following is incorrect?
(1) Cellular organization and consciousness are defining properties of living things.
(2) Systematics involves taxonomy along with phylogeny.
(3) Trinomial nomenclature system was proposed by Caesalpino and Bauhin.
(4) Scientific names of cultivated plants have been standardized through ICNCP
650. Which one of the following is defining property of living being?
(1) Growth (2) Reproduction (3) Metabolism (4) Unconsciousness
651. Find the wrong statement.
(1) Classification is the process of grouping of anything into categories based on some easily observable characters.
(2) Dogs, cats, animals represent taxa at different levels.
(3) Each category commonly represent a taxon.
(4) Rank of taxon does not represent unit of classification.
652. According to five-kingdom system of classification, how many kingdoms contain multicellular eukaryotes?
(1) Three kingdom (2) Four kingdom
(3) Two kingdom (4) All the five kingdoms
653. Mango belongs to class dicot of angiosperms and belongs to order:
(1) Poales. (2) Sapindales. (3) Rannales. (4) Parietales.
654. According to Whittaker Actinomycetes are included in kingdom:
(1) Monera. (2) Protista. (3) Fungi (4) Slime moulds.
655. Taxonomy that is based of sequencing of DNA and chemical nature of proteins is:
(1) Chemotaxonomy. (2) Cytotaxonomy.
(3) Adansonian taxonomy. (4) Karyotaxonomy.
656. Rattus rattus scientific name is an example of:
(1) Autonyms. (2) Tautonyms. (3) Synonyms. (4) Homonyms.
657. While preparing herbarium dried specimens are poisoned by using a solution of:
(1) $10\%HgCl_2$. (2) $0.1\%Mg_2Cl_2$. (3) $0.1\%CuSO_4$. (4) $0.1\%HgCl_2$.
658. Binomial epithet includes:
(1) Genetic epithet + author citation
(2) Specific epithet
(3) Generic epithet + specific citation
(4) Generic epithet + specific epithet + author citation
659. Select correct statement with respect to binomial nomenclature.
(1) Species name starts with capital letter when it denotes a person or common name
(2) Biological names are generally in italies and written in Latin
(3) Handwritten names are separately underlined to indicate their latin origin
(4) In compound specific name, the second word denotes variety or sub-species
660. "The Darwin of the 20^{th} century" is associated with or known as:
(1) Nominalistic concept of species. (2) Biological concept of species.
(3) Triple crown in biology. (4) More than one option is correct.
661. The taxonomical aid which is an index of plant species in an area is:
(1) Grade. (2) Manuals. (3) Monographs. (4) Flora.
662. Mark the incorrect one (with respect to taxonomic keys).
(1) Based on the contrasting in nature.
(2) Generally analytical in nature.

(3) East statement is called a lead.
(4) Same taxonomic key can be used for different taxonomic category.

663. Suffix used for class is:
(1) -ini. (2) -idae. (3) -opsida. (4) -eae.

664. According to the three-kingdom classification, the fungi are placed in:
(1) Animalia. (2) Protista. (3) Plantae. (4) Monera.

665. Garden having the main collection of the woody species is known as:
(1) Arboretum. (2) Bambusetum. (3) Pinetum. (4) ' Orchidarium.

666. First true phylogenetic system on classification was given by:
(1) Eicher. (2) Engler and Prantl. (3) de Jussieu. (4) de Candole.

667. In India, NBRI is situated at:
(1) Calcutta. (3) Madras. (2) Bombay. (4) Lucknow.

668. According to ICBN, the name of the class ends with suffix:
(1) -Phyceae. (2) -Opsida. (3) -Eae. (4) Both (1) and (2).

669. In the hierarchial classification, the number of obligate categories is:
(1) 7 . (2) 8 . (3) 6 . (4) 12 .

670. Multicellular, eukaryotic consumers belong to kingdom.
(1) Fungi. (2) Protista. (3) Plantae. (4) Animalia.

671. A taxonomic system based on all phenotypic similarities, equally weighted, and with regard to evolutionary relationship is called:
(1) Phylogeny: (2) Cladistics. (3) Classical taxonomy. (4) Phenetics.

672. Taxon and category differ in:
(1) Taxon is recognized and assigned, whereas category is abstract.
(2) Taxon is a group of real organisms, whereas category is a rank or level in a hierarchy.
(3) Both (1) and (2).
(4) None of these.

673. Four-kingdom system arose with the distinction of:
(1) Animalia and planatae. (2) Invertebrate and vertebrate.
(3) Prokaryotes and eukaryotes. (4) Autotrophs and heterotrophs.

674. ICBN stands for:
(1) International classification of biological nomenclature.
(2) International class of biological nomenclature.
(3) International code of botanical nomenclature.
(4) International classification of biological naming.

675. "Taxa" differs from "taxon" due to:
(1) This being a higher taxonomic category than taxon.
(2) This being lower taxonomic category than taxon.
(3) This being the plural of taxon.
(4) This being the singular of taxon.

676. The total number of species that are known and described range between:
(1) 0.5 and 1.0 million (2) 1.1 and 1.2 million
(3) 1.7 and 1.8 million (4) 2.5 and 3.0 million

677. Which of the following combinations is correct for wheat?
(1) Genus Triticum, Family Anacardiaceae, Order Poales, Class Monocotyledonae
(2) Genus Triticum, Family Poaceae, Ordier Poales, Class Dicotyledonae
(3) Gemis Triticum, Family Poaceae, Order Sapindales, Class Monocotyledonae
(4) Genus Triticum, Family Poaceae, Order Poales, Class Monocotyledonae

678. Classification of organization based on evolutionary as well as genetic relationship is called:
(1) Biosystematics. (2) Phenetics. (3) Numerical taxonomy. (4) Cladistics.

679. Founder of binomial nomenclature was:
(1) Linnaeus. (2) Mendel. (3) Darwin. (4) Lamarck.

680. Two apparently similar populations are intersterile. They are member of:
(1) One species. (2) Two biospecies.
(3) Two sibling species. (4) Two parapatric species.

681. The phylogenetic system refers to grouping:
(1) According to floral similarities.
(2) Of plants in order of their increasing complexities.
(3) According to evolutionary trends.
(4) According to all morphological characters.

682. Interbreeding is possible between two members of:
(1) Order. (2) Family. (3) Genus. (4) Species.

683. A place of collection of dried plant specimen is:
(1) Arborateum. (2) Herbarium. (3) Botanical gardens. (4) All of these.

684. The term "Systematic" was given by:
(1) Lamarck. (2) Linnaeus. (3) Candolle. (4) Endlicher.

685. "Pinetum" is a:
(1) Garden with collection of conifers. (2) Collection of books on conifers.
(3) Collection of seeds of conifers. (4) All of these.

686. Who proposed the concept of phylogeny?
(1) Linnaeus (2) Darwin (3) Haeckel (4) Ernst Mayr

687. Match the following column-I and column-II (with respect to term given by).

Column-I	Column -II
Taxon	(i) Eichler
Phylum	(ii) Lamarck
Division	(iii) Adolf Meyer
Phylogeny	(iv) Cuvier

(1) 1 (iii), 2 (iv), 3 (ii), 4 (i) (2) 1 (iii), 2 (iv), 3 (i), 4 (ii)
(3) 1 (iv), 2 (iii), 3 (ii), 4 (i) (4) 1 (i), 2 (ii), 3 (iii), 4 (iv)

688. The category that includes related species is:
(1) Class. (2) Order. (3) Division. (4) Genera.

689. Find the odd one (with respect to category).
(1) Algae (2) Moss (3) Fern (4) Conifer

690. Which taxonomy establishes the numerical degree of relationship among individuals?
(1) Adansonian taxonomy ` (2) Experiment taxonomy
(3) Turill taxonomy (4) Karyotaxonomy

691. Carl Woese found that the six kingdoms naturally cluster into three main domains on the basis of the sequence of:
(1) $16s - r$ RNA. (2) $28s - r$ RNA.
(3) 23 s − rRNA. (4) 5.8 s − rRNA.

692. Match column A and B
(A) Species concepts
(B) Given by

1.	Static species concept	(i) G Simpson
2.	Dynamic species concept	(ii) Linnaeus

3. Biological species concept (iii) Lamarck
4. Ecological species (iv) E Mayr

(1) 1 (ii), 2 (i), 3 (iii), 4 (iv) (2) 1 (iii), 2 (ii), 3 (i), 4 (iv)
(3) 1 (ii), 2 (iii), 3 (iv), 4 (i) (4) 1 (ii), 2 (i), 3 (iv), 4 (iii)

693. Find odd one out with respect to taxonomic category.
(1) Poaceae (2) Homonidae (3) Malvaceae (4) Insecta

694. Find an incorrect statement with respect to taxonomical aids.
(1) Herbarium serves as quick referral system in taxonomical studies.
(2) Plant species are grown for identification purpose in botanical gardens.
(3) Keys are based on the pair of contrasting characters called couplet.
(4) Monographs provide a complete index of species in a given area.

695. Find out if one of the following categories is incorrect with respect to Mango.
(1) Family – Anacardiaceae (2) Order - Polemoniales
(3) Class – Dicotyledonae (4) Division – Angiospermae

696. A group of freely interbreeding plants constitutes a:
(1) Genera. (2) Species. (3) Class. (4) Family.

697. Standard size of a herbarium sheet is:
(1) 40×32 cm (2) 38×29 cm (3) 41×29 cm (4) 42×26 cm

698. Nomenclature of cultivated plants are standardized through:
(1) ICBN. (2) ICNB. (3) ICNCP. (4) ICZN.

699. Which scientist published his classification in the book "Families of Flowering Plants"?
(1) Hutchinson (2) Theophrastus (3) John Ray (4) Bentham and Hooker

700. Biological concept of species was proposed by:
(1) Linnaeus. (2) Julian Huxley. (3) Meyer. (4) John Ray.

701. The international code of Botanical nomenclature was adopted in:
(1) 1955 . (2) 1961 . (3) 1964. (4) 1968.

702. The book "Historia Generalis Plantarum" was written by:
(1) John Ray. (2) Takhtajan. (3) Darwin. (4) Lamarck.

703. Find the odd one out (with respect to category).
(1) Division (2) Tribe (3) Order (4) Class

704. Match the following
Species are static, constant, fixed and immutable (i) Linnaeus
Concept of phylogeny (ii) Lamarck and Darwin
Species are dynamic and (iii) Ernst Mayr
(iv) Haeckel liable to undergo change with the passage of time

(1) 1 (i), 2 (iv), 3 (ii), 4 (iii) (2) 1 (iii), 2 (i), 3 (iv), 4 (ii)
(3) 1 (ii), 2 (iv), 3 (iii), 4 (i) (4) 1 (iv), 2 (ii), 3 (i), 4 (iii)

705. The complete compilation of information about any taxon/ genus/species is called:
(1) Manual. (2) Monograph. (3) Flora. (4) Catalogue.

706. The taxonomic category of pteridophytes is (with respect to suffix):
(1) Order. (2) Class. (3) Division. (4) Family.

707. In taxonomic keys, the pairs of contrasting traits are known as:
(1) Doublet. (2) Couplet. (3) Lead. (4) Alleles.

708. Mark the incorrect one:
(1) Growth is extrinsic type in nonliving objects.
(2) Self-consciousness is present in human beings.
(3) Most living and some nonliving objects are capable of reproduction.
(4) Cellular organization is the defining feature of life forms.

709. Members of which of the following groups are not concerned with the same taxon?

(1) Rice, Dog. (2) Insects, Housefly

(3) Potato, Mango (4) Man, Wheat

710. Mark the odd one (with respect to category).

(1) Orchid (2) Tortoise (3) Palm (4) Eucalyptus

711. Find the odd one with respect to taxon.

(1) Gourds (2) Palms (3) Conifers (4) Grasses

712. Find the odd one (with respect to taxonomic categories).

(1) Muscidae (2) Primata (3) Sapindales (4) Poales

713. According to Botanical nomenclature which is not possible?

(1) Tautonyms (2) Synonyms (3) Bionyms (4) Autonyms

714. The term "New systematics" was introduced by:

(1) Linnaeus. (2) AP Candolle. (3) Julian Huxley. (4) Bentham and Hooker.

In Dinomial nomenclature, every organism has:

(1) Two names-one Latin and other common.

(2) Two names-one scientific and other common.

(3) Two names by two scientists.

(4) One scientific name with two words-generic and specific.

715. Taxon is:

(1) Any ope of taxonomic grouping like species, family, phylum based on similarity of traits.

(2) A rank in hierarchical classification.

(3) A group of closely related families.

(4) A group of closely related organisms.

716. Which of the following is species?

(1) Chordata (3) Carnivora (2) Mammalia (4) Canis familaris

717. Choose the correct sequence of taxonomic categories.

(1) Class - Phylum - Order - Family - Genus Species

(2) Division - Class - Order - Family - Genus - Species

(3) Division - Class - Family - Order - Genus - Species

(4) Phylum - Order - Class - Family - Genus - Species

718. Holotype is:

(1) Specimen cited with original description.

(2) Specimen used by an author as nomenclature type.

(3) Duplicate of first identified specimen.

(4) None of these.

719. Phenetic classification of organisms is based on:

(1) Sexual characteristics.

(2) Observable characteristics of existing organisms.

(3) The ancestral lineage of existing organisms.

(4) Dendrogram based on DNA characteristics.

720. In the light of recent classification of living organisms into three domains of life (bacteria, archaea, and eukarya), which one of the following statements is true about archaea?

(1) Archaea resemble eukarya in all respects.

(2) Archaea have some novel features that are absent in other prokaryotes and eukaryotes.

(3) Archaea completely differ from both prokaryotes and eukaryotes.

(4) Archaea completely differ from prokaryotes.

721. "Taxa" differs from "taxon" due to:

(1) This being a higher taxonomic category than taxon.

(2) This being lower taxonomic category than taxon.
(3) This being the plural of taxon.
(4) This being the singular of taxon.

722. A group of related genera, with still less number of similarities as compared to the genus and species, constitutes.
(1) Order. (2) Class. (3) Family. (4) Division.

723. Actinomycetes are being put under which of the following?
(1) Kingdom Fungi (2) Kingdom Monera
(3) Kingdom Plantae (4) Kingdom Protoctista

The living World

724. The biological definition of a species depends on
(1) Geographical distribution of two groups of organisms.
(2) Reproductive isolation of two groups of organisms.
(3) Anatomical and developmental differences between the two groups of organisms.
(4) Difference in the adaptation of two groups of organisms.

725. The biological concept of species was formulated by:
(1) Mayr. (2) Stebbins. (3) Heywood. (4) Love.

726. National Botanical Research Institute is situated at:
(1) Lucknow. (2) Kolkata. (3) Mumbai. (4) Chennai.

727. Drosophila pseudoobscura and Drosophila persimilis are good examples of:
(1) Sympatric species. (2) Allopatric species.
(3) Polytypic species. (4) Sibling species.

728. Binomial system of nomenclature was given by:
(1) Julian Huxley. (2) Bentham and Hooker.
(3) Linnaeus. (4) Casper Bauhin.

729. The number of species classified in "Species Plantarum" is:
(1) 5000 . (2) 6000 . (3) 4000 . (4) 3800 .

730. New systematics introduced by Sir Julian Huxley is also called:
(1) Phenetics. (2) Cladistics. (3) Biosystematics. (4) Numerical taxonomy.

731. Phenetics is also called:
(1) Classical taxonomy. (2) Artificial taxonomy.
(3) Neotaxonomy. (4) Adansonian taxonomy.

732. Term "Alfa taxonomy" was given by:
(1) Turril. (2) Bateson. (3) Linnaeus. (4) Darwin.

733. Phenetic classification of organisms is based on:
(1) Sexual characteristics.
(2) Observable characteristics of existing organisms.
(3) The ancestral lineage of existing organisms.
(4) Dendrogram based on DNA characteristics.

734. Linnaean system of plant classification is based on:
(1) Morphological and anatomical characters. (2) Evolutionary trends.
(3) Floral characters. (4) None of the above.

735. Systematics term was given by:
(1) Lamarck. (2) de Candolle. (3) Linnaeus. (4) Darwin.

736. The phylogenetic system of classification was put forth by:
(1) Carolus Linnaeus. (2) George Bentham and Joseph Dalton Hooker.
(3) Aristotle. (4) Adolf Engler and Karl Prantl.

737. Which one of the taxonomic aids can give comprehensive account of complete compiled information of any one genus or family at a particular time?
(1) Taxonomic key (2) Flora (3) Herbarium (4) Monograph

738. Systematic botany means:
(1) System analysis.
(2) Systematic arrangement of organs of plants.
(3) Systematic study of organelles and tissues.
(4) Methodical study of plants, dealing with identification, naming and classification.

739. Datura innoxia belongs to the order and family, respectively:
(1) Malvales, Fabaceae. (2) Passiflorales, Poaceae.
(3) Polemoniales, Solanaceae. (4) Sapindales, Anacardiaceae.

740. A phylum common to unicellular animals and plants is:
(1) Monera. (2) Plantae. (3) Fungi. (4) Protista.

741. The most widespread group of organisms of earth belongs to kingdom:
(1) Monera. (2) Protista. (3) Fungi. (4) Plantae.

742. Classification that considers several characters of ancestors is:
(1) Phylogenetic. (2) Artificial. (3) Natural. (4) Phyllotaxy.

743. According to Whittaker's five kingdom classification the unicellular, nonnucleated organisms are placed in:
(1) Monera. (2) Protista. (3) Plantae. (4) Animalia.

744. Which of the following statements regarding universal rules of nomenclature is wrong?
(1) The first word in a biological name represents the genus.
(2) The first word denoting the genus starts with a capital letter.
(3) Both the words in a biological name, when handwritten are separately underlined.
(4) Biological names are generally in Greek and written in italics.

745. "Systema Naturae" was written by:
(1) Ernst Mayr. (2) Carolus Linnaeus.
(3) RH Whittaker. (4) WM Stanley.

746. Which of the following domain was not suggested by Woese?
(1) Animalia (3) Archaea (2) Bacteria (4) Eukarya

747. Which of the following combinations is correct for maize?
(1) Genus : Zea, Family : Anacardiaceae,. Order: Poales, Class: Monocotyledoneae
(2) Genus : Zea, Family : Poaceae, Order : Poales, Class : Dicotyledoneae
(3) Genus : Zea, Family : Poaceae, Order : Sapindales, Class : Monocotyledoneae
(4) Genus : Zea, Family : Poaceae, Order: Poales, Class: Monocotyledoneae

748. IUCN stands for:
(1) Indian Union for Conservation of Nature.
(2) International Union for Conservation of Nature.
(3) Indian Union for Chemical Nomenclature.
(4) International Union for Conservation of Nutrients.

749. A group of related genera, with still less number of similarities as compared to the genus and species, constitutes:
(1) Order. (2) Class. (3) Family. (4) Division.

750. Classification of organisms based on evolutionary as well as genetic relationships is called:
(1) Biosystematics. (2) Phenetics. (3) Numerical taxonomy. (4) Cladistics.

751. True species are:
(1) Interbreeding. (2) Sharing the same niche.
(3) Feeding on the same food. (4) Reproductively isolated.

752. Nomenclature given by Linnaeus is:

(1) Binomial. (2) Trinomial. (3) Phylogenetic. (4) Natural.

753. Taxonomic hierarchy refers to:

(1) Stepwise arrangement of all categories for classification of plants and animals.

(2) A group of senior taxonomists, who decide the nomenclature of plants and animals.

(3) A list of botanists or zoologists, who have worked on taxonomy of a species or group.

(4) Classification of a species based on fossil record.

754. Father of Botany is:

(1) Aristotle. (3) Darwin. (2) Robert Hooke. (4) Theophrastus.

755. Family is placed between:

(1) Order and Class. (2) Genus and Species.

(3) Class and Genus. (4) Order and Genus.

756. The framework system of classification in which various taxonomic categories are arranged in order of logical sequence is called:

(1) Systematics. (2) Classification. (3) Hierarchy. (4) Taxon.

757. The basic unit of classification is:

(8) Species. (2) Genus. (3) Family. (4) Phylum.

758. The term "Taxonomy" was introduced by:

(1) de Candolle. (2) Bentham and Hooker. (3) Linnaeus. (4) Huxley.

759. Scientific study of diversity of organisms and their evolutionary relationships is:

(1) Morphology. (2) Anatomy. (3) Taxonomy. (4) Systematics.

760. Arrange the following in the ascending order of Linnaean hierarchy.

(1) Kingdom-Order-Species-Genus-Class-FamilyPhylum

(2) Kingdom-Family-Genus-Order

(3) Kingdom-Phylum-Class-Order-Family-GenusSpecies

(4) Species-Genus-Family-Order-Class-PhylumKingdom

761. Select the wrong statements.

I. Lower the taxon, more are the characteristics that the members within the taxon share.

II. Order is the assemblage of genera which exhibit a few similar characters.

III. Cat and dog are included in the same family-Felidae.

IV. Binomial nomenclature was introduced by Carolus Linnaeus.

(1) I, II, and III (2) II, III, and IV (3) I and IV (4) II and III

762. Animals are classified into hierarchical group. In which one of the following, the largest number of species are found?

(1) Genus (2) Order (3) Family (4) Class

763. Which one of the following is not a correct statement?

(1) Herbarium houses dried, pressed and preserved plant specimens.

(2) Botanical gardens have collection of living plants for reference.

(3) A museum has collection of photographs of plants and animals.

(4) Key is a taxonomic aid for identification of specimens